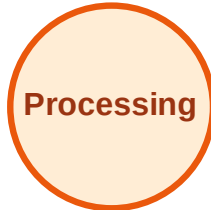


BFS Traversal

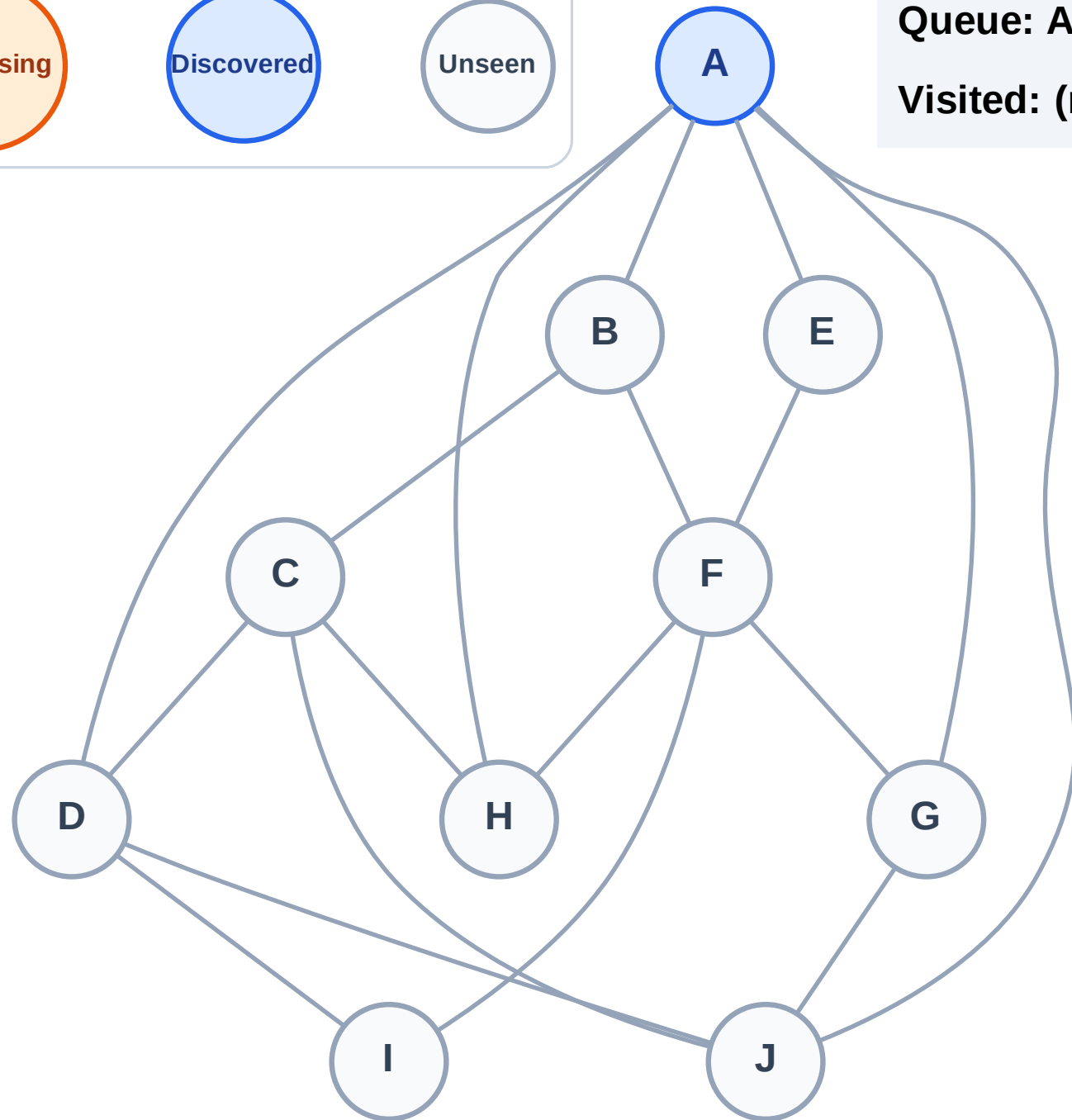
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

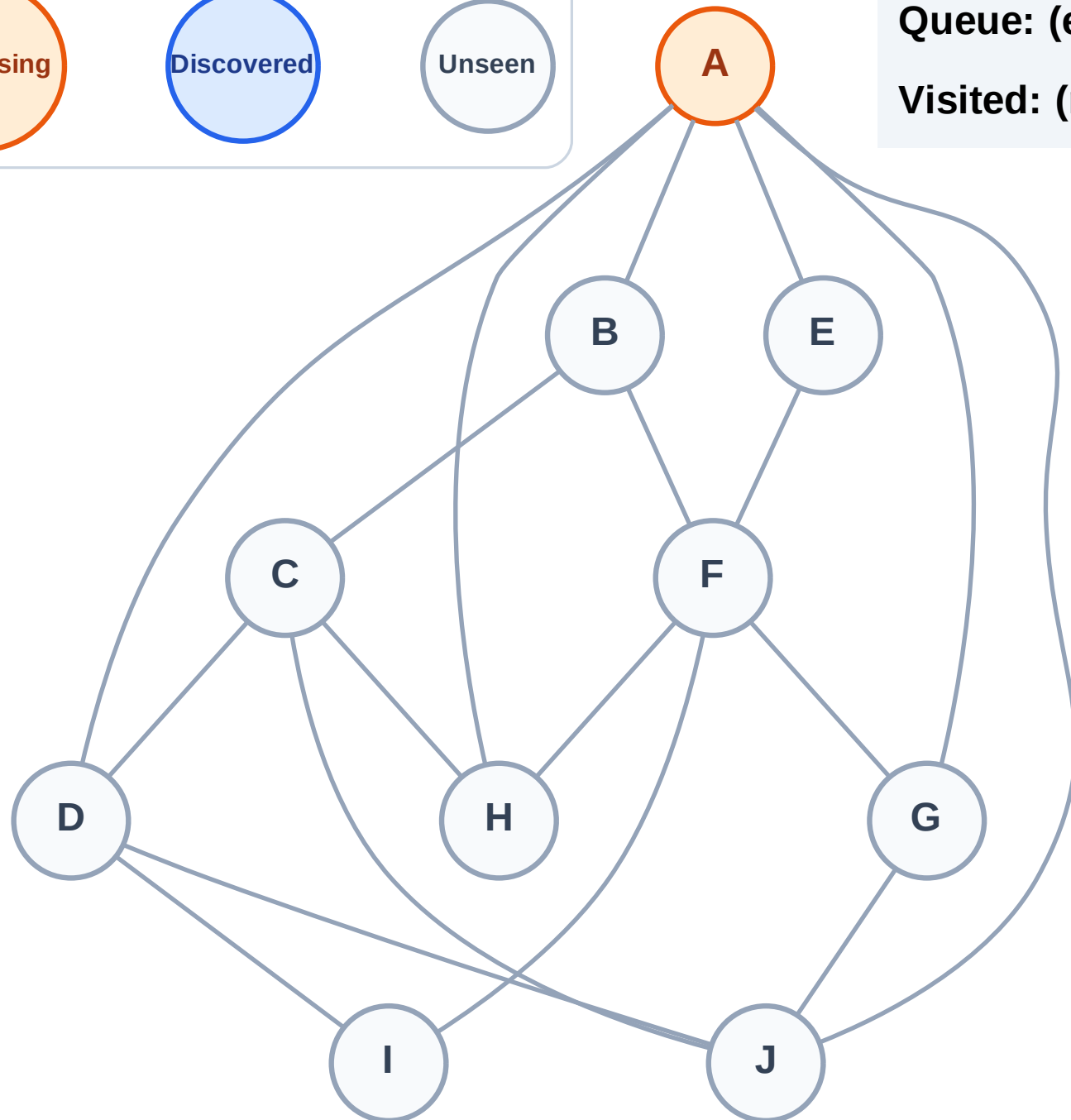
Step 2: Process node A

Legend



Queue: (empty)

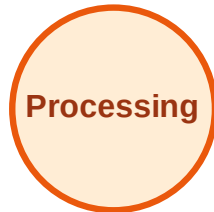
Visited: (none)



BFS Traversal

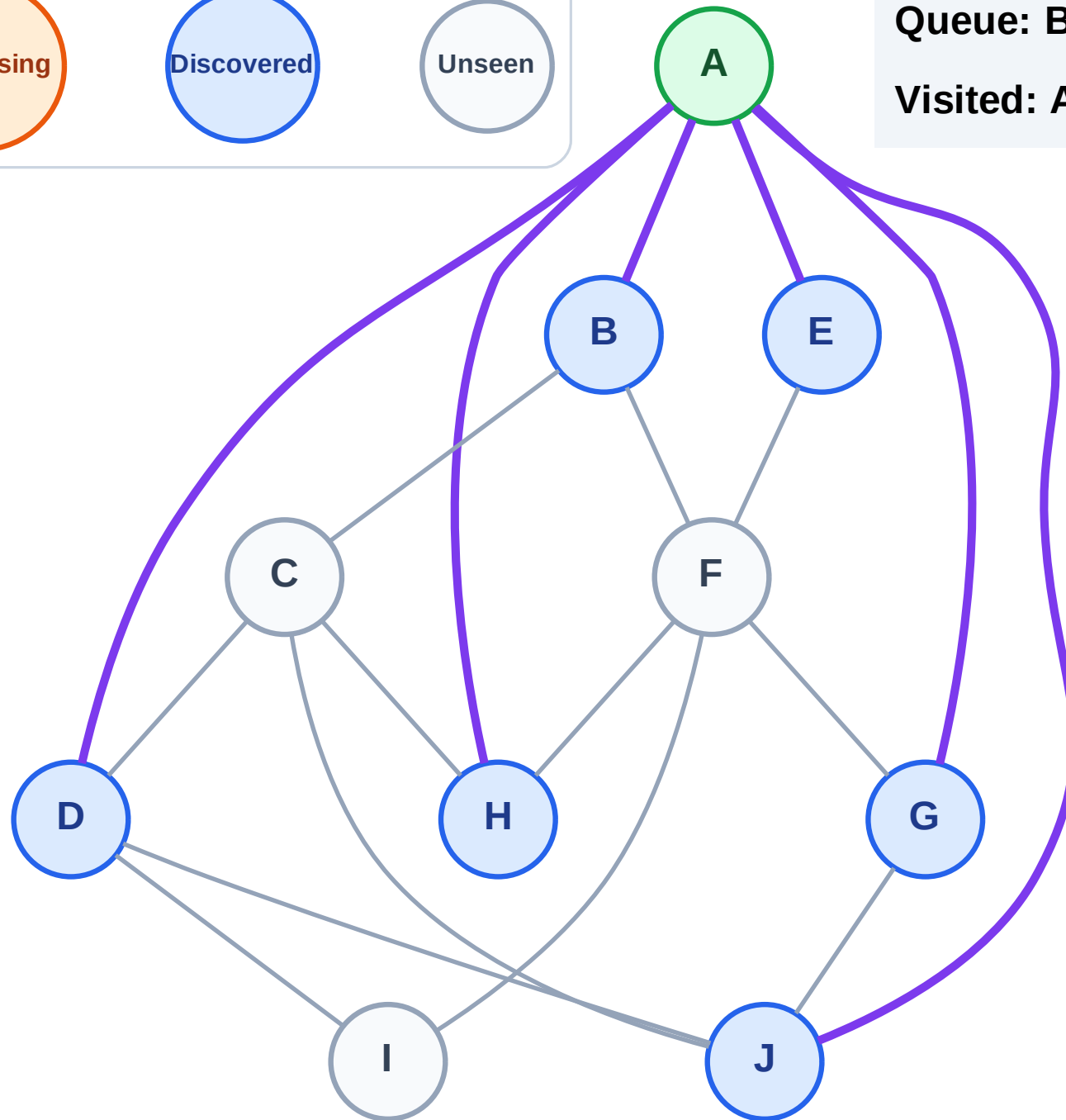
Step 3: Discover B, D, E, G, H, J from A

Legend



Queue: B → D → E → G → H → J

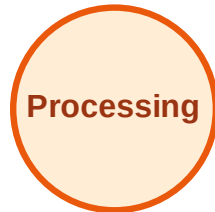
Visited: A



BFS Traversal

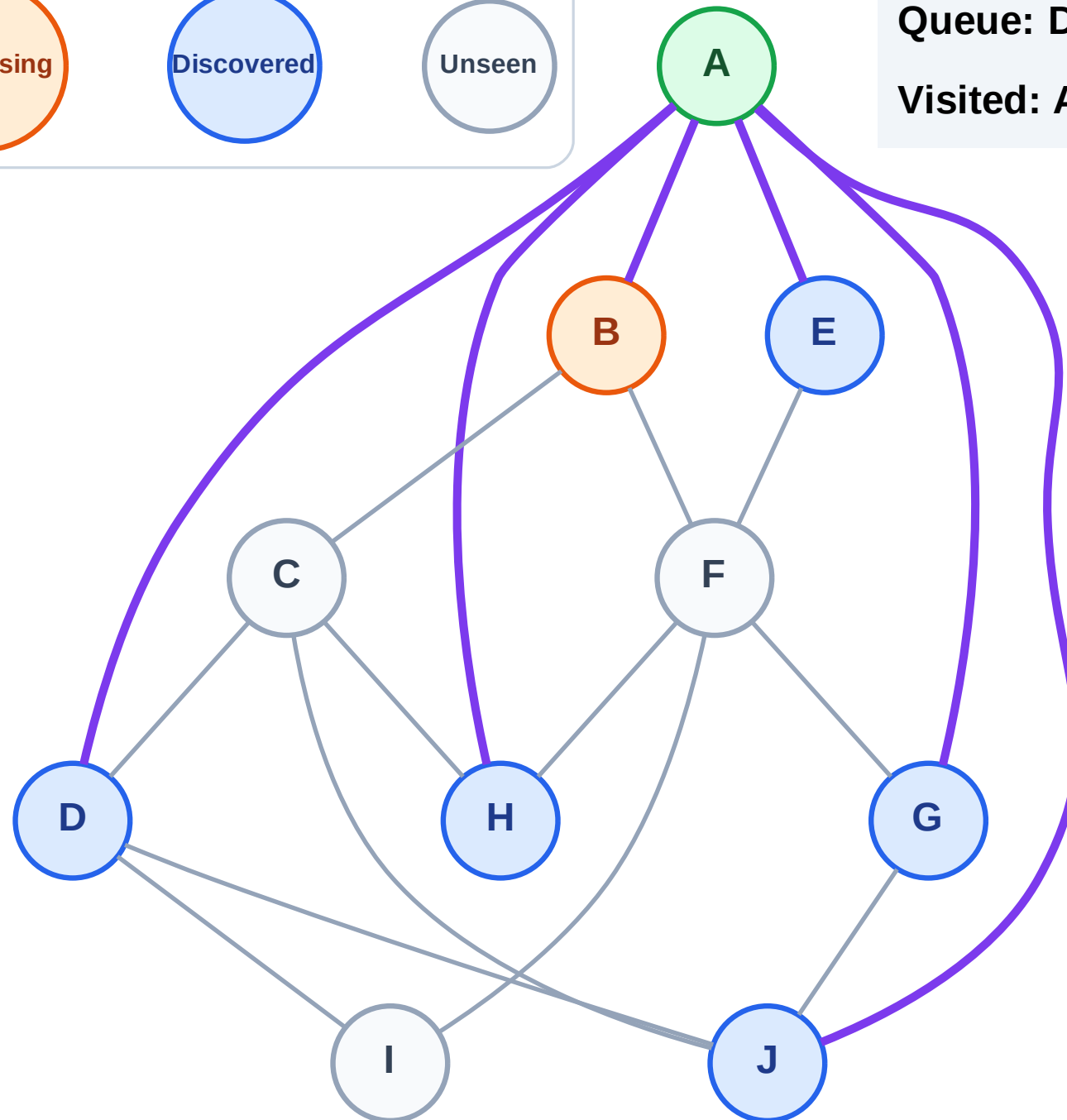
Step 4: Process node B

Legend



Queue: D → E → G → H → J

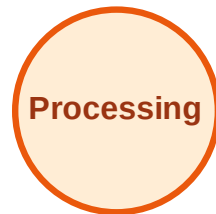
Visited: A



BFS Traversal

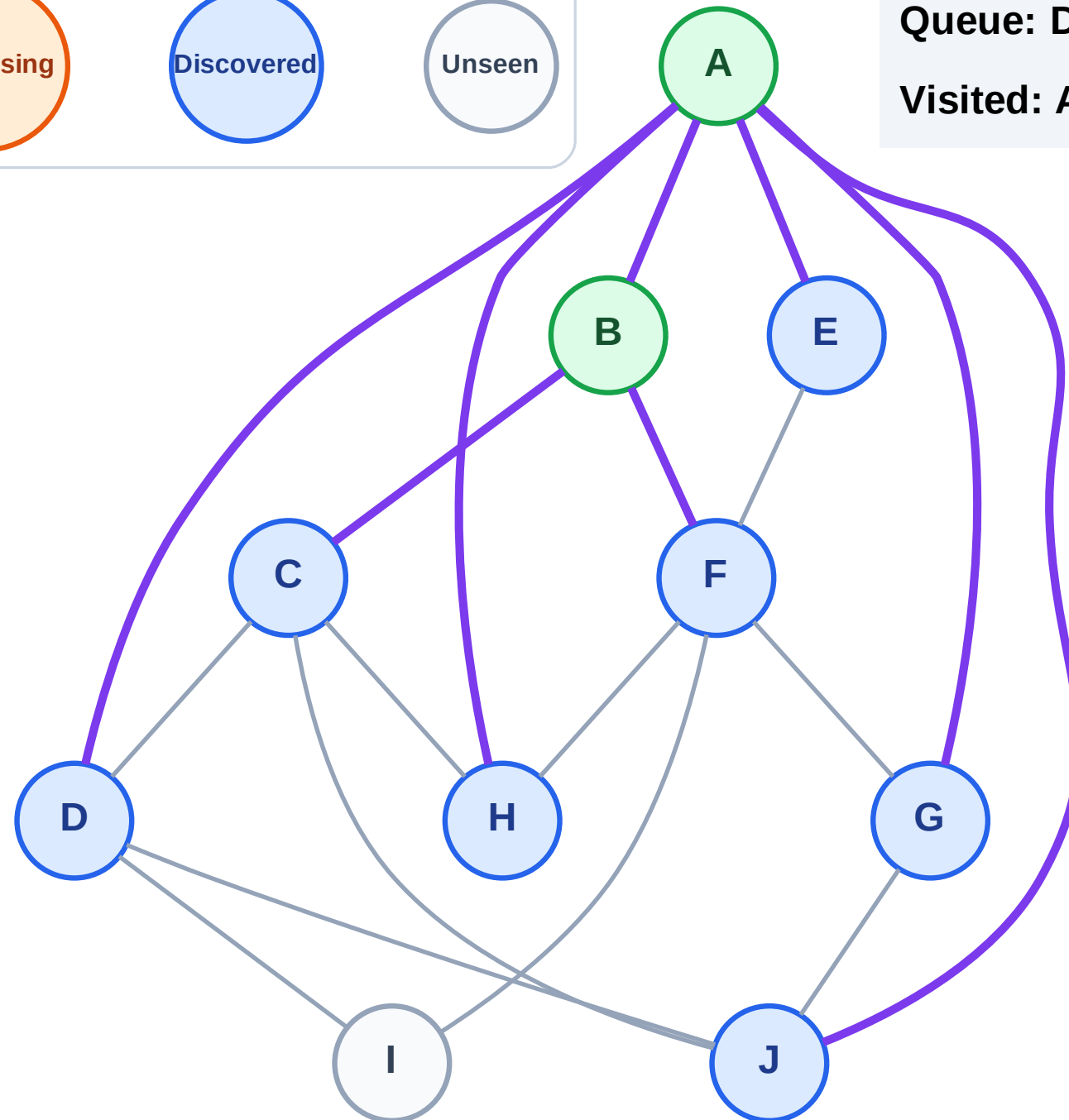
Step 5: Discover C, F from B

Legend



Queue: D → E → G → H → J → C → F

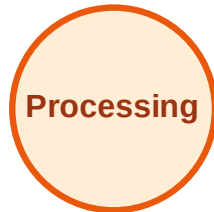
Visited: A, B



BFS Traversal

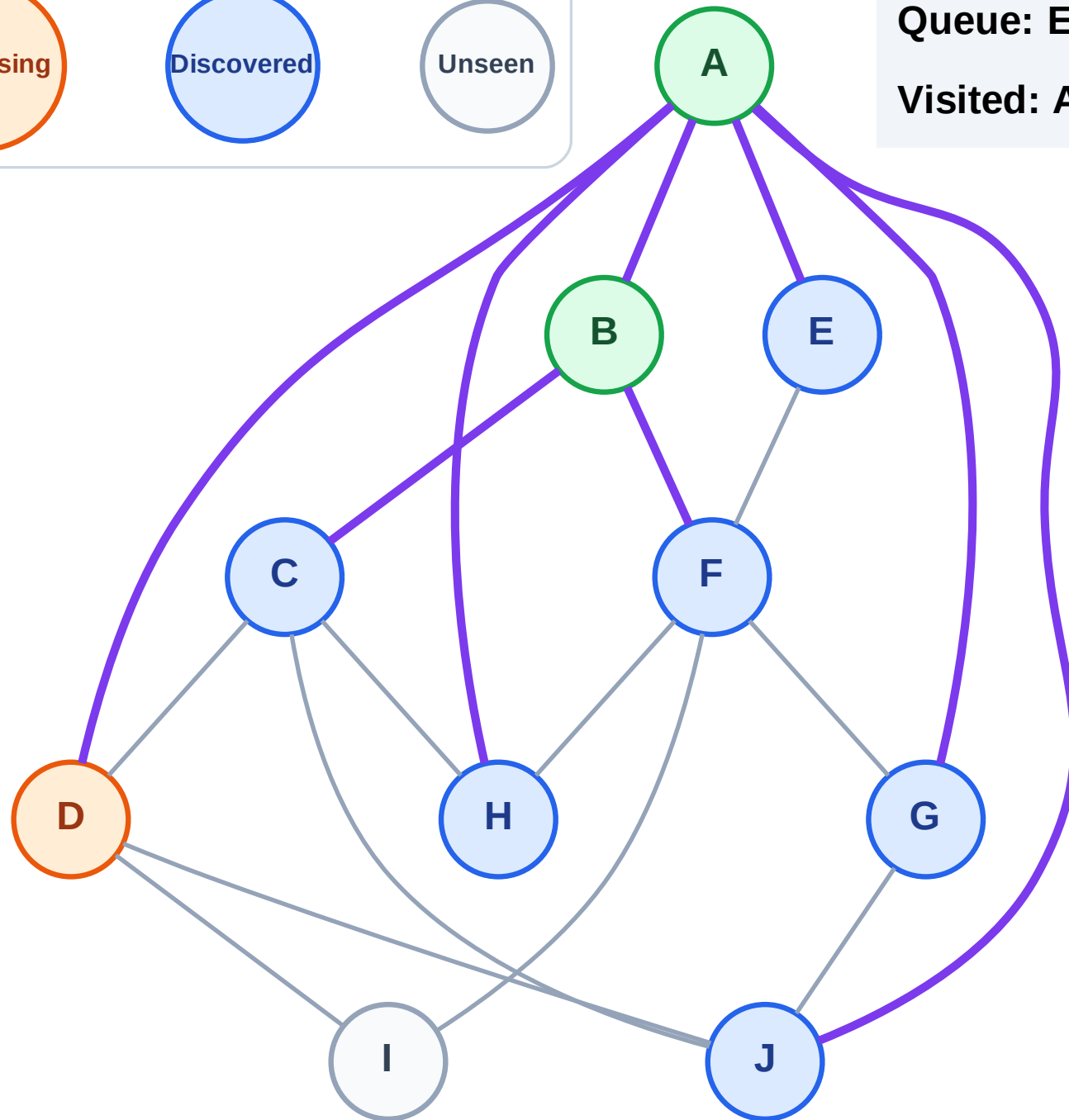
Step 6: Process node D

Legend



Queue: E → G → H → J → C → F

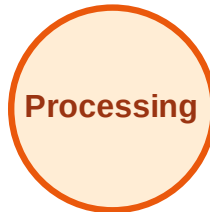
Visited: A, B



BFS Traversal

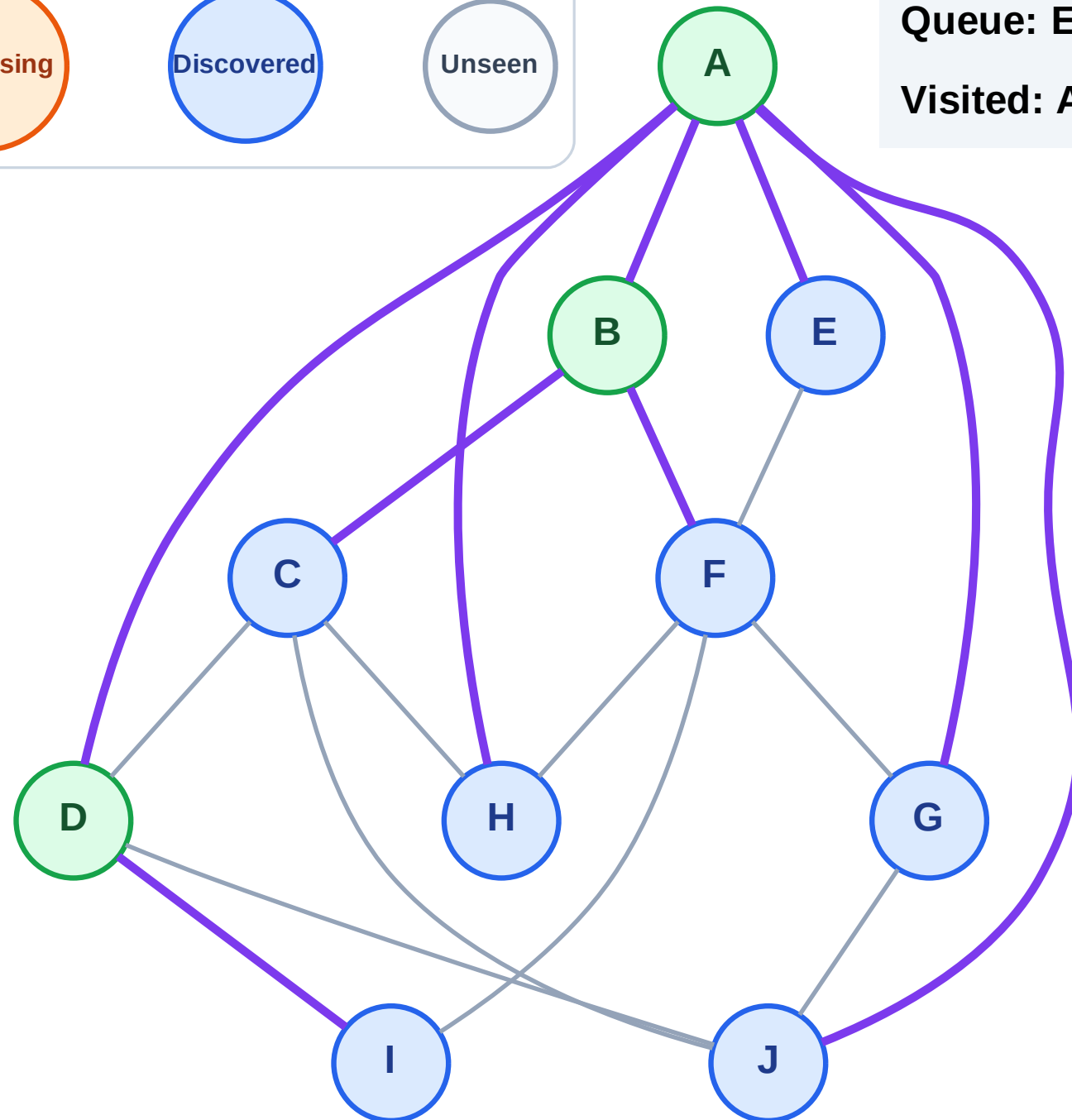
Step 7: Discover I from D

Legend



Queue: E → G → H → J → C → F → I

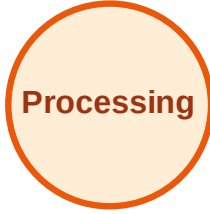
Visited: A, B, D



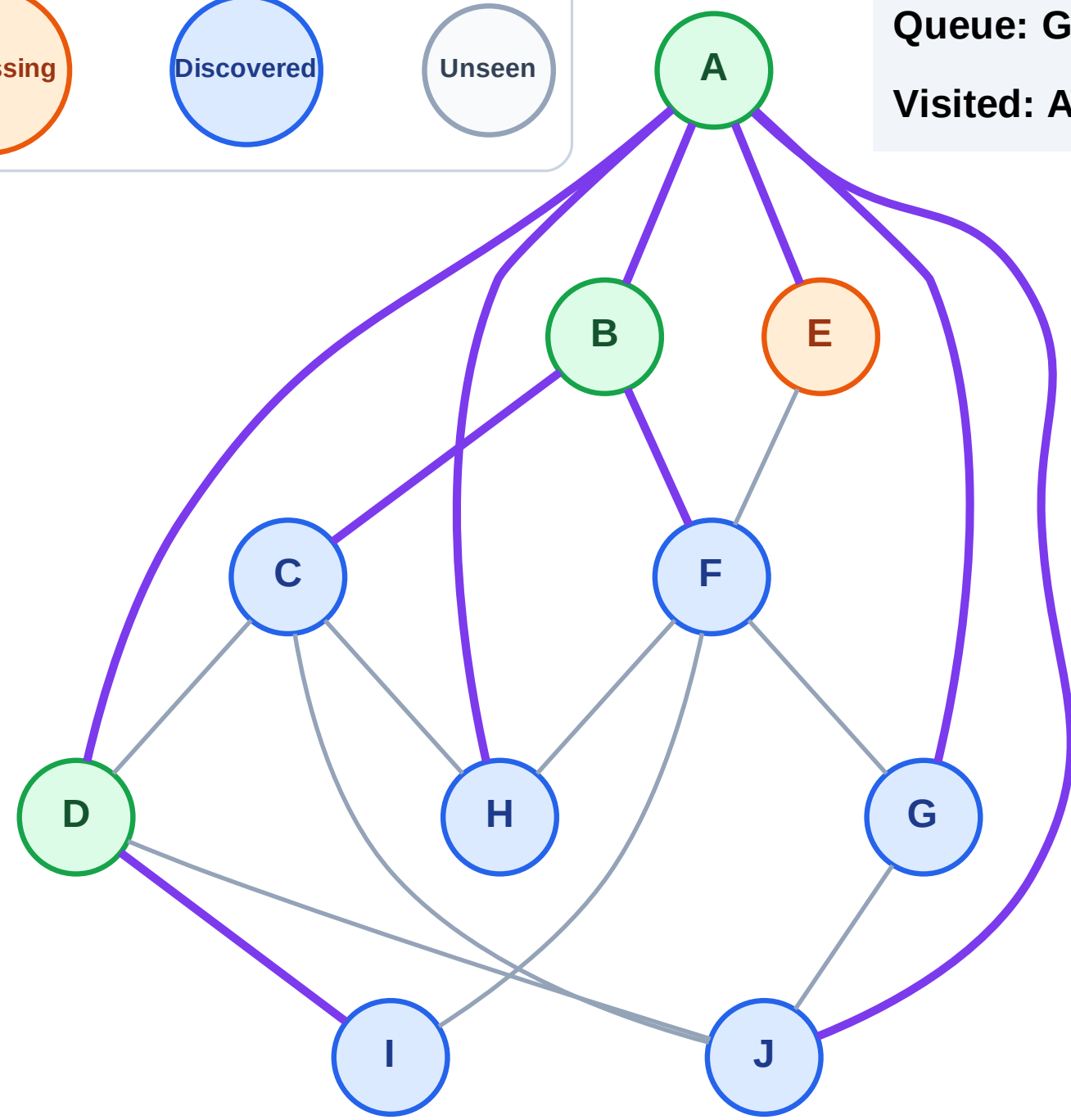
BFS Traversal

Step 8: Process node E

Legend



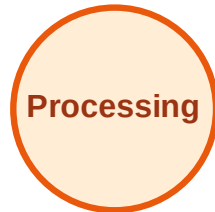
Queue: G → H → J → C → F → I
Visited: A, B, D



BFS Traversal

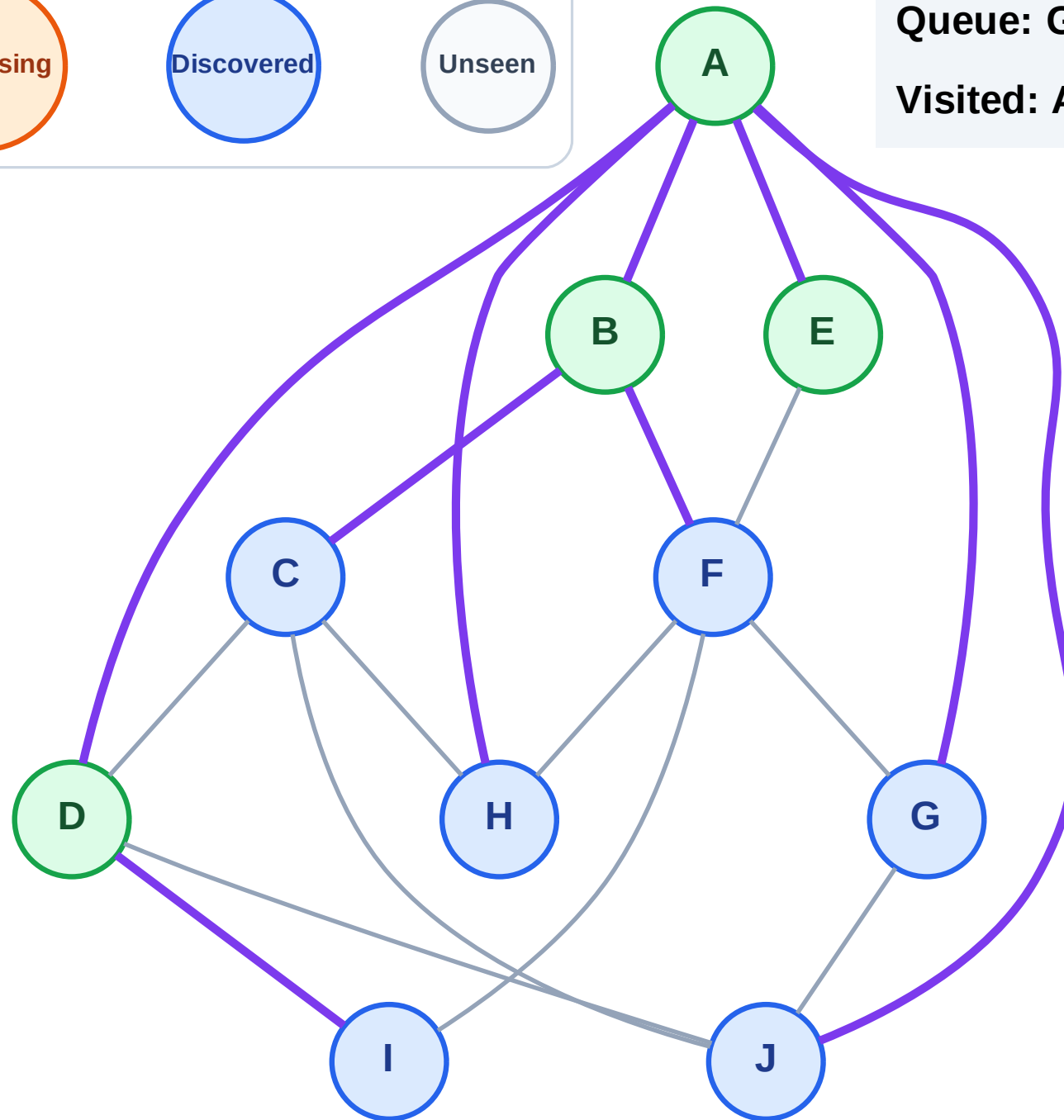
Step 9: No new neighbors from E

Legend



Queue: G → H → J → C → F → I

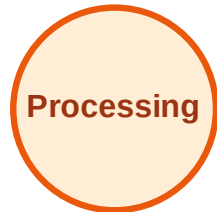
Visited: A, B, D, E



BFS Traversal

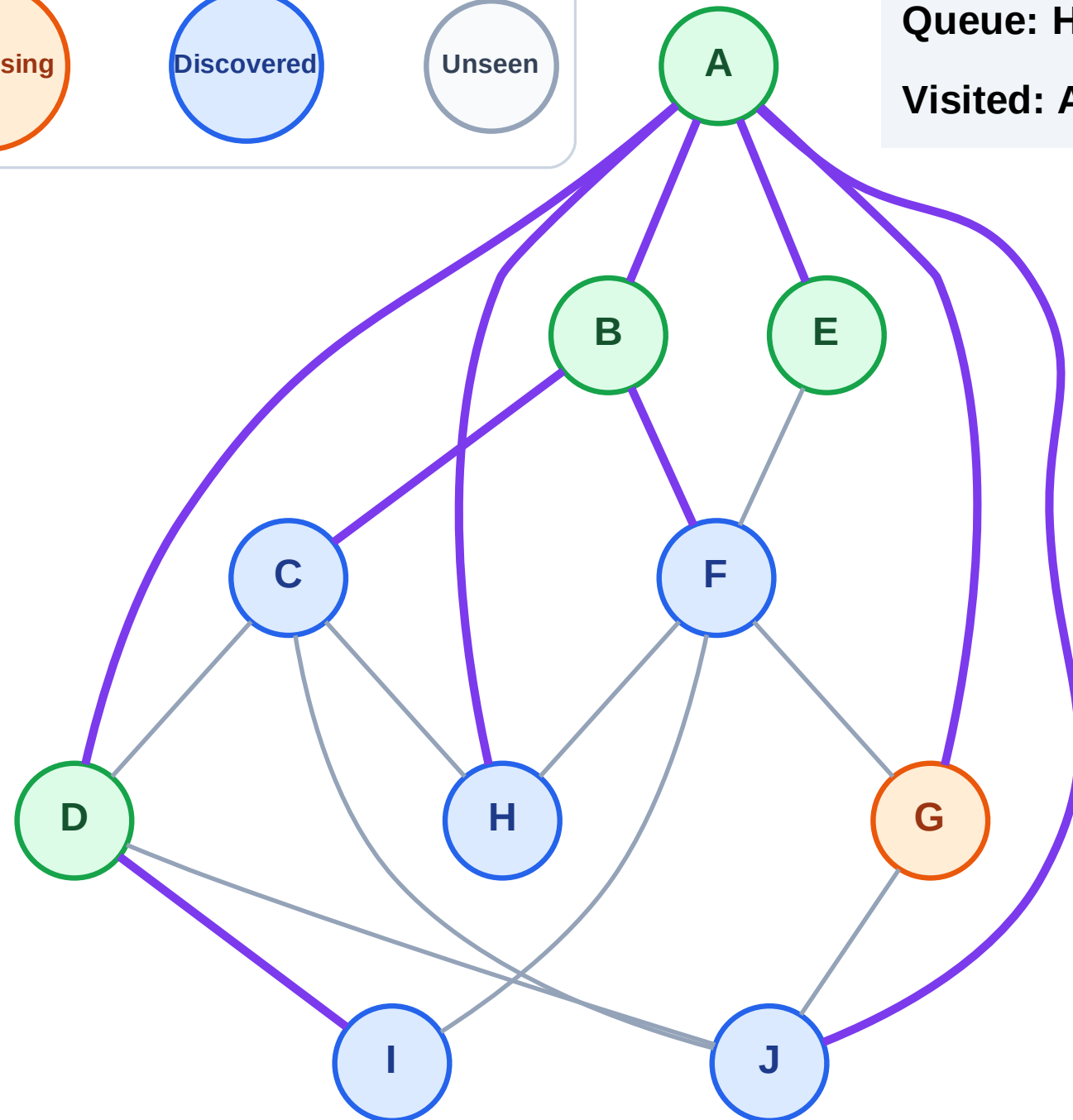
Step 10: Process node G

Legend



Queue: H → J → C → F → I

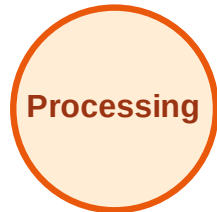
Visited: A, B, D, E



BFS Traversal

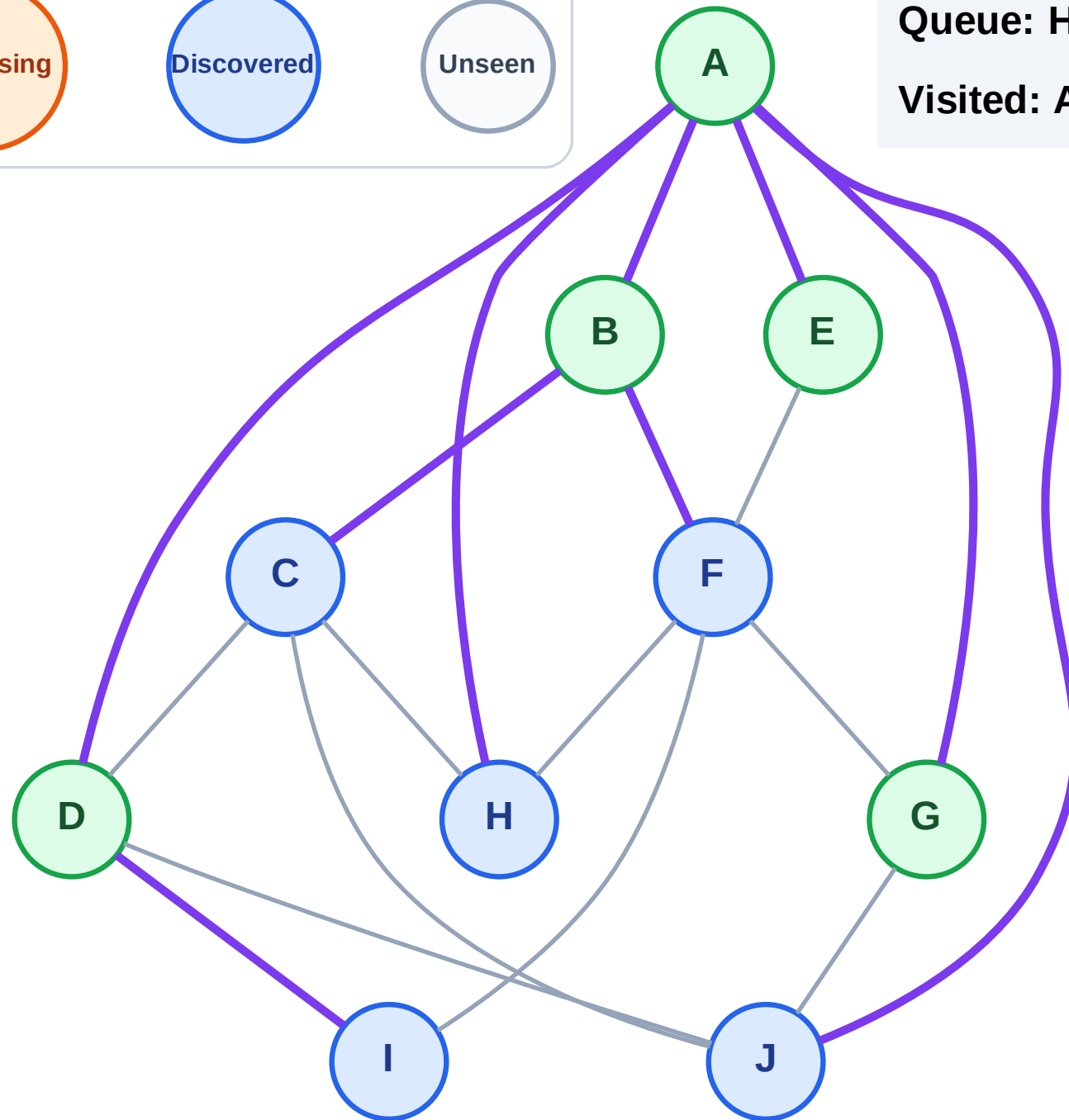
Step 11: No new neighbors from G

Legend



Queue: H → J → C → F → I

Visited: A, B, D, E, G



BFS Traversal

Step 12: Process node H

Legend

Complete

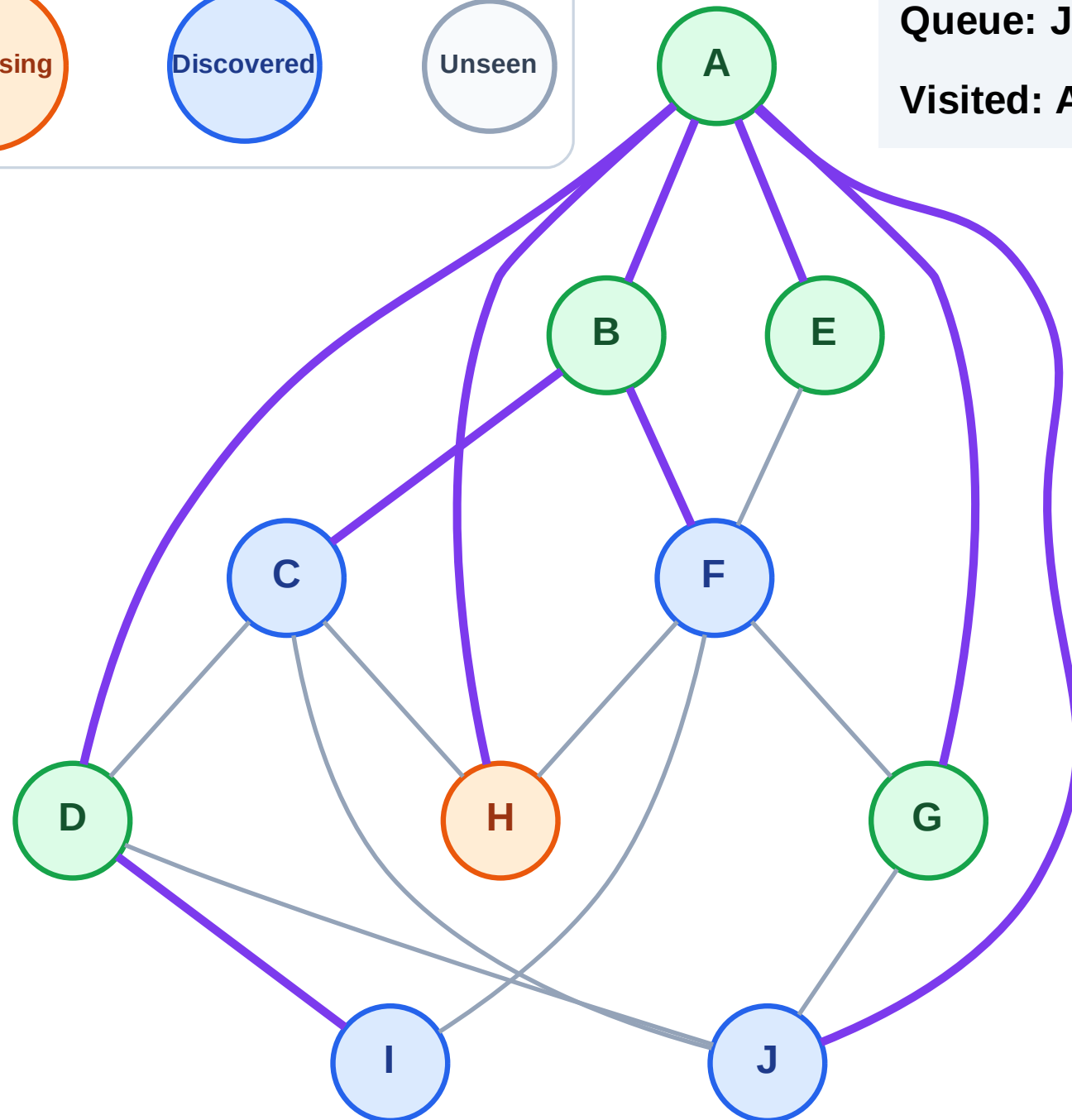
Processing

Discovered

Unseen

Queue: J → C → F → I

Visited: A, B, D, E, G



BFS Traversal

Step 13: No new neighbors from H

Legend

Complete

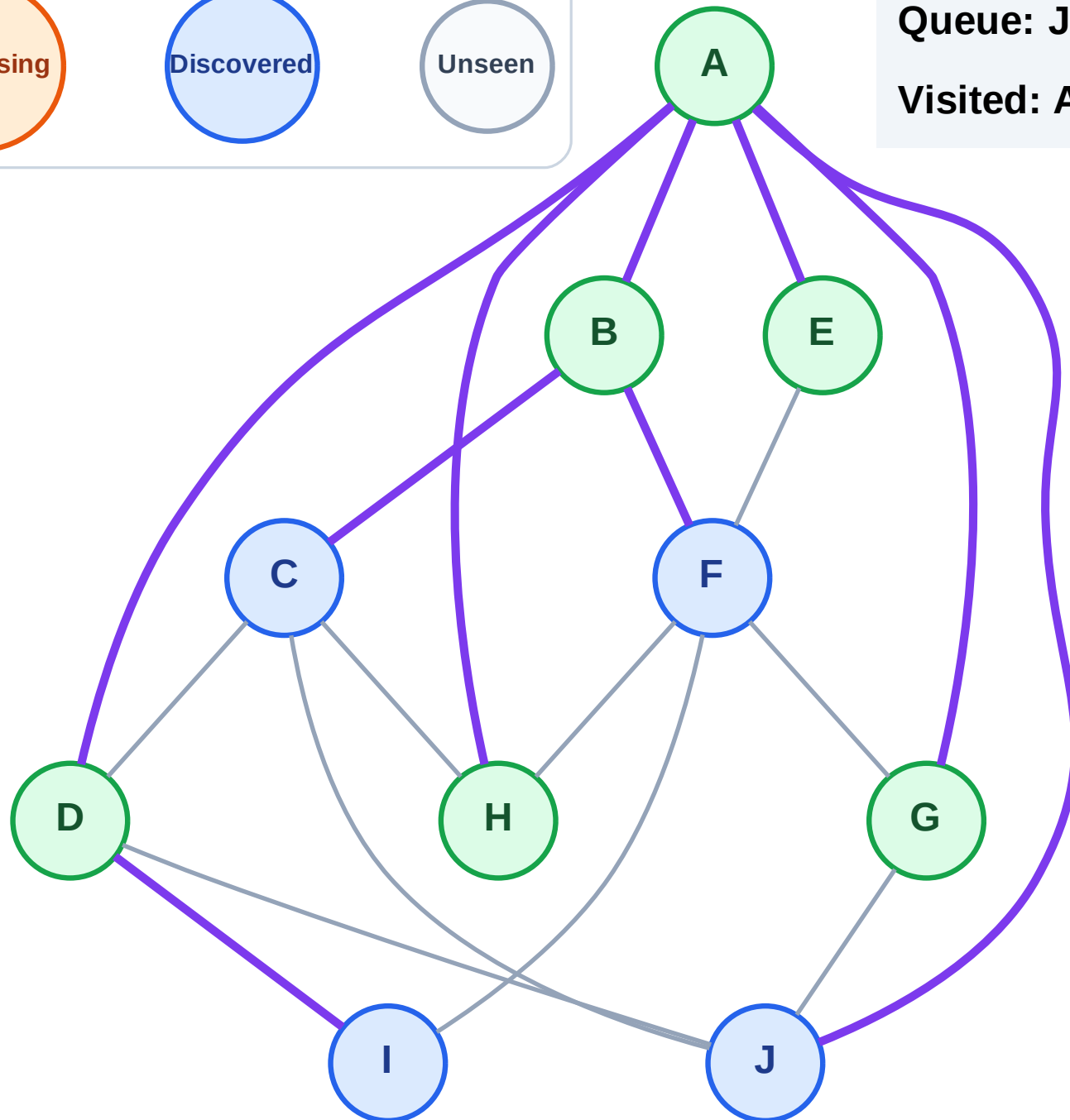
Processing

Discovered

Unseen

Queue: J → C → F → I

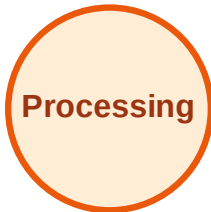
Visited: A, B, D, E, G, H



BFS Traversal

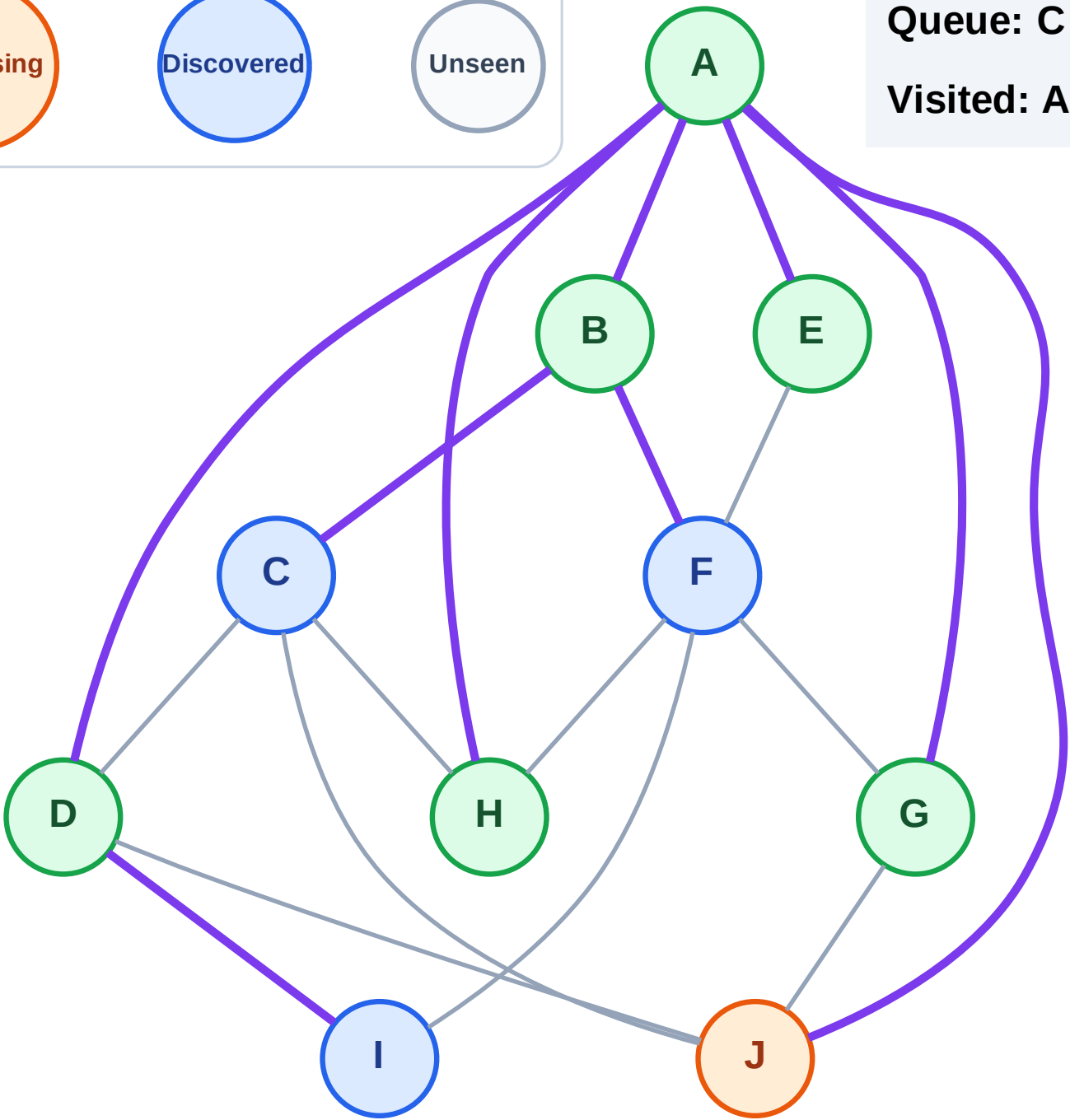
Step 14: Process node J

Legend



Queue: C → F → I

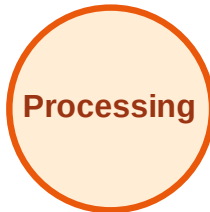
Visited: A, B, D, E, G, H



BFS Traversal

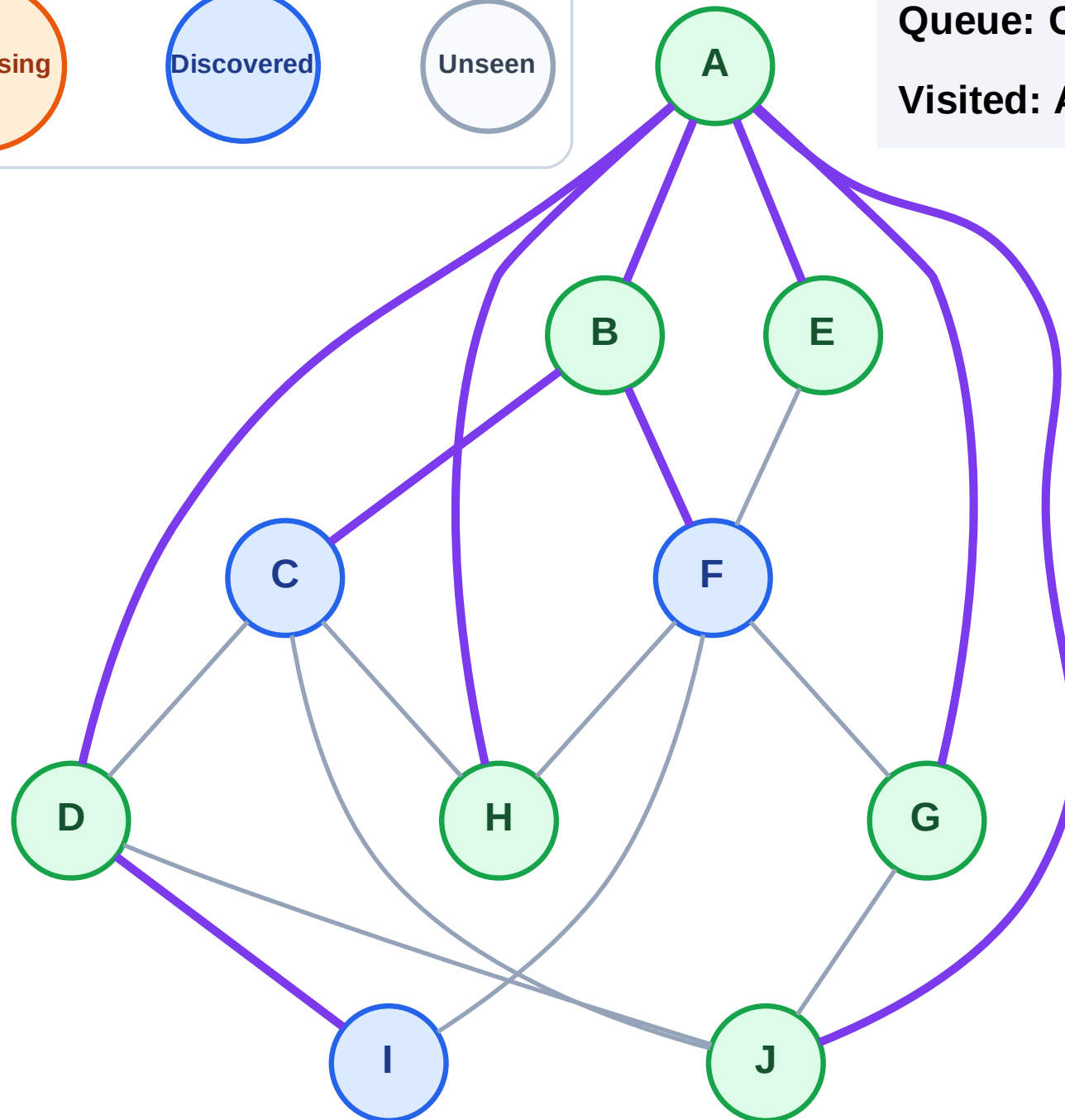
Step 15: No new neighbors from J

Legend



Queue: C → F → I

Visited: A, B, D, E, G, H, J



BFS Traversal

Step 16: Process node C

Legend

Complete

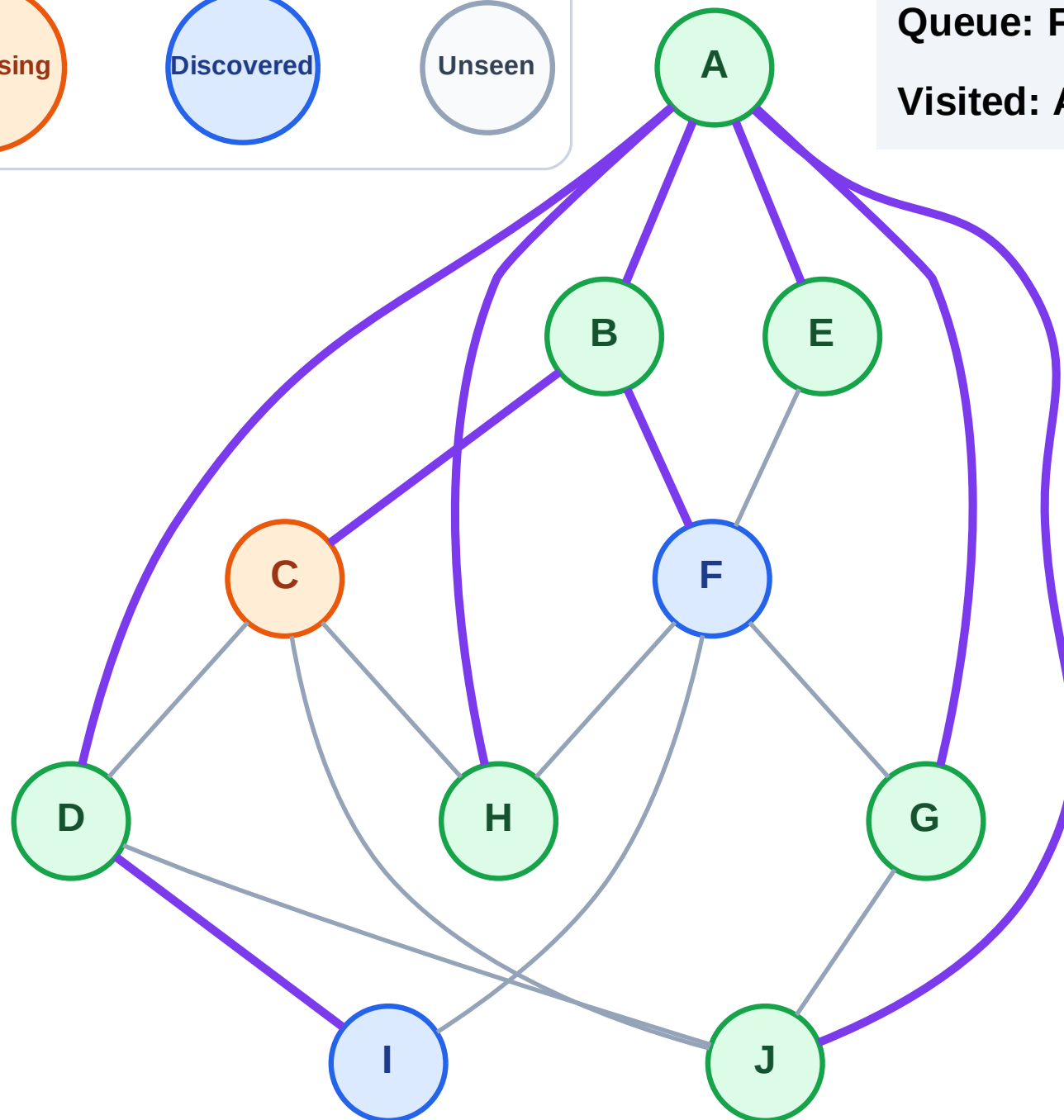
Processing

Discovered

Unseen

Queue: F → I

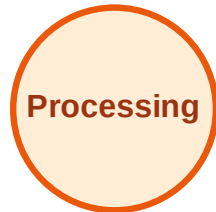
Visited: A, B, D, E, G, H, J



BFS Traversal

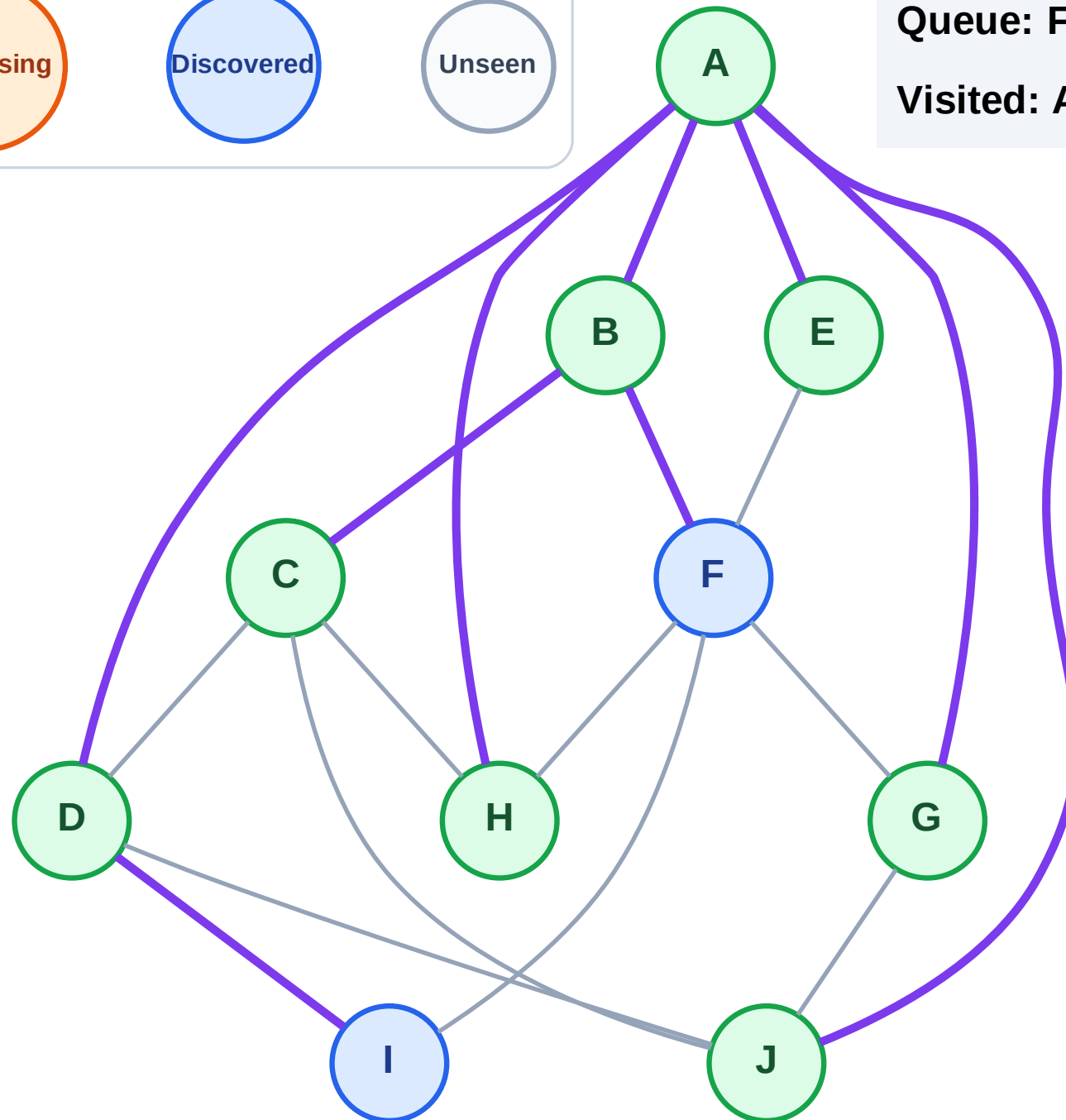
Step 17: No new neighbors from C

Legend



Queue: F → I

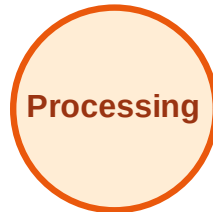
Visited: A, B, C, D, E, G, H, J



BFS Traversal

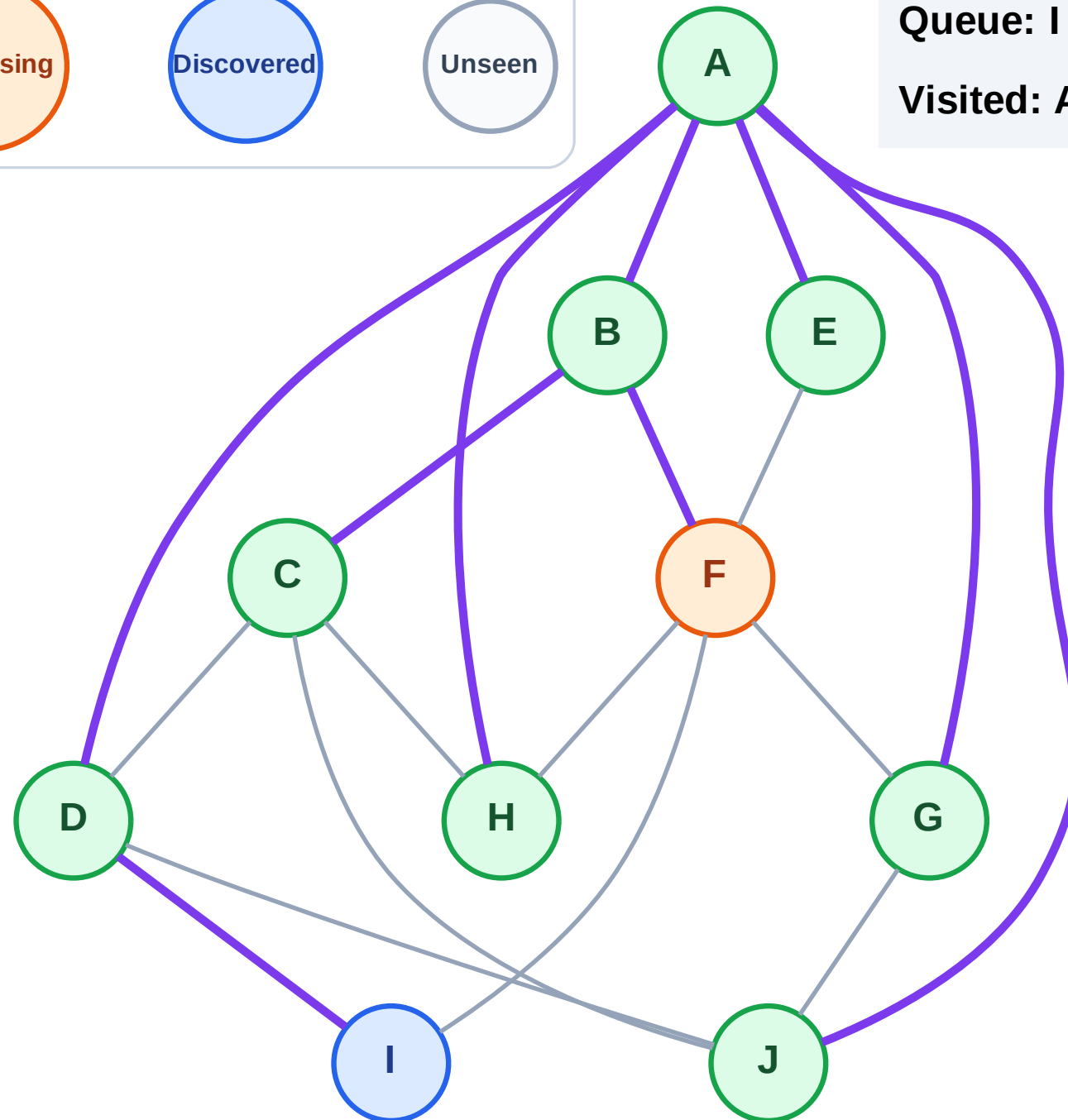
Step 18: Process node F

Legend



Queue: I

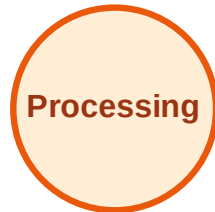
Visited: A, B, C, D, E, G, H, J



BFS Traversal

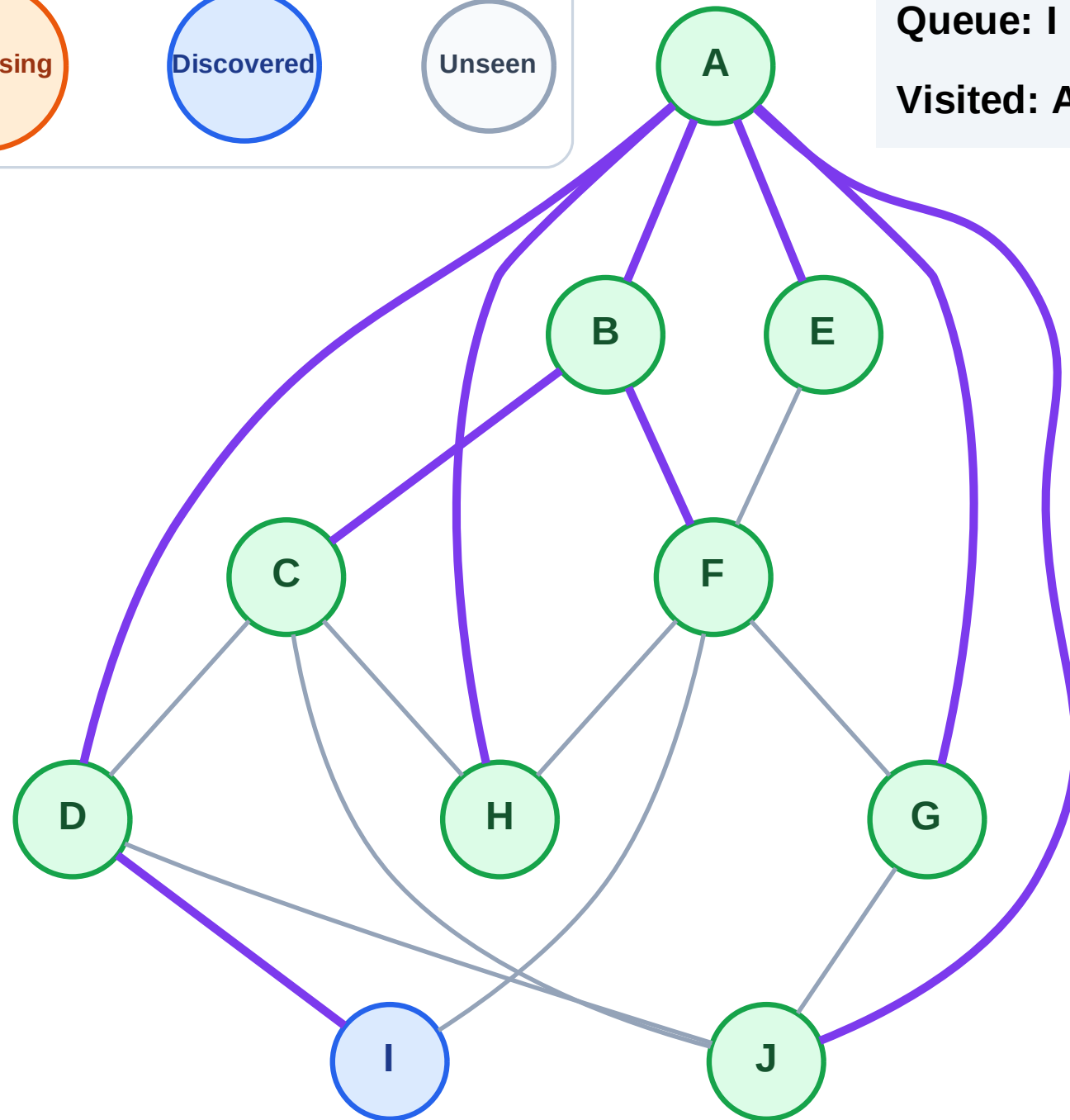
Step 19: No new neighbors from F

Legend



Queue: I

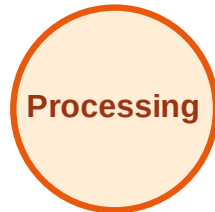
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

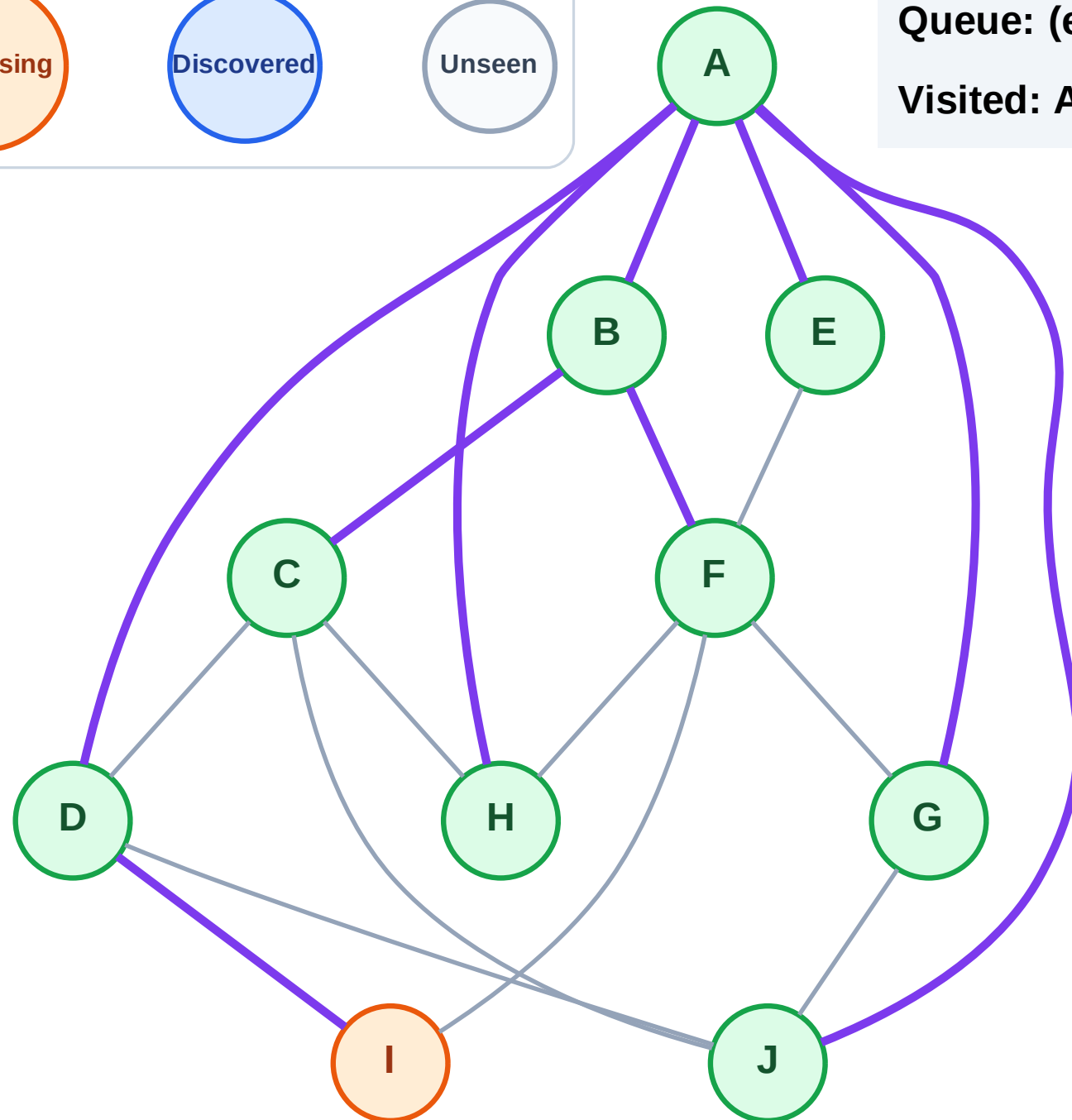
Step 20: Process node I

Legend



Queue: (empty)

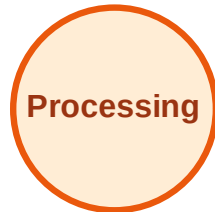
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

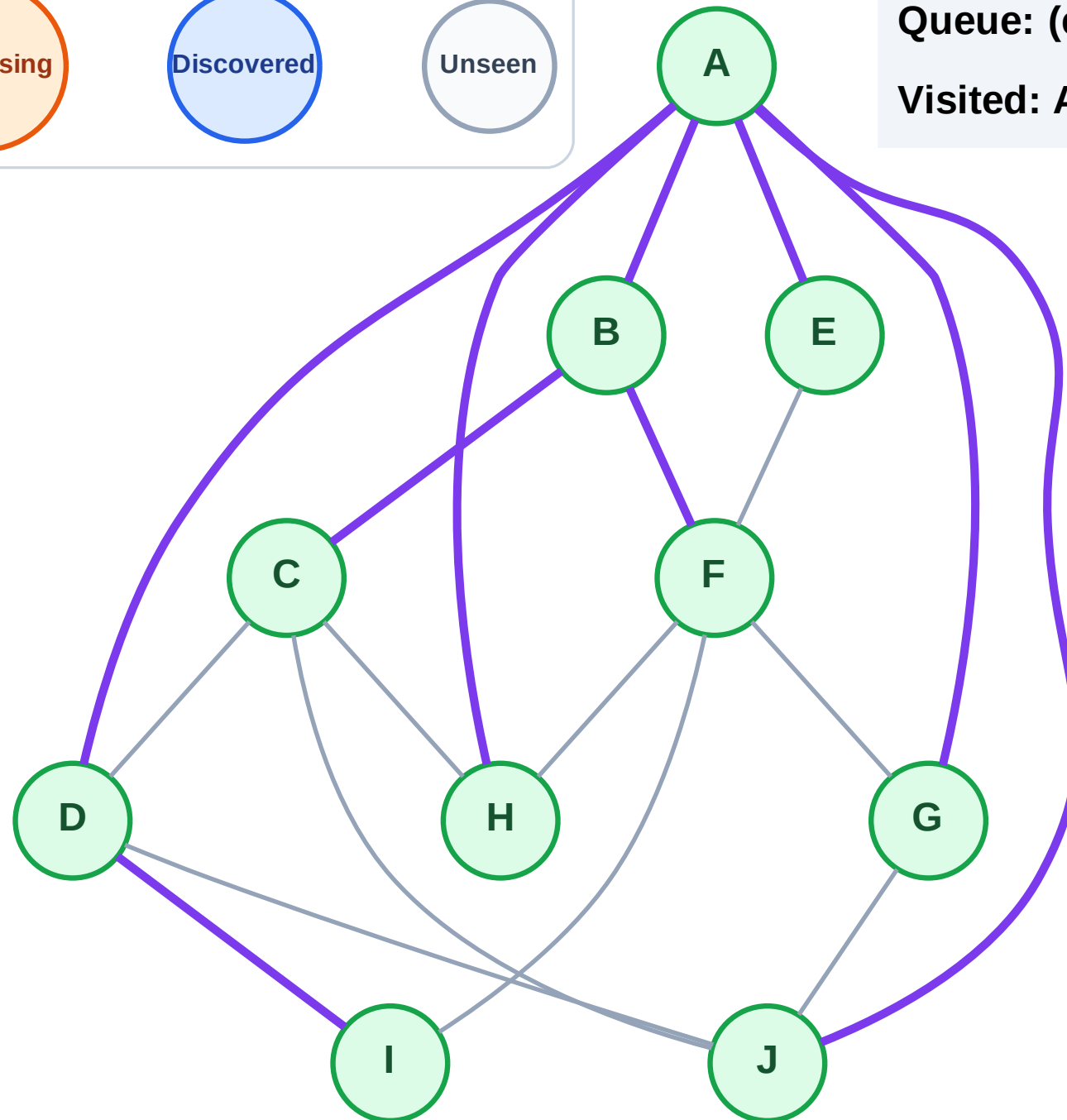
Step 21: No new neighbors from I

Legend



Queue: (empty)

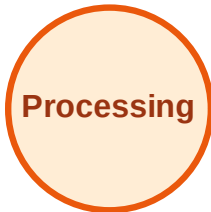
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

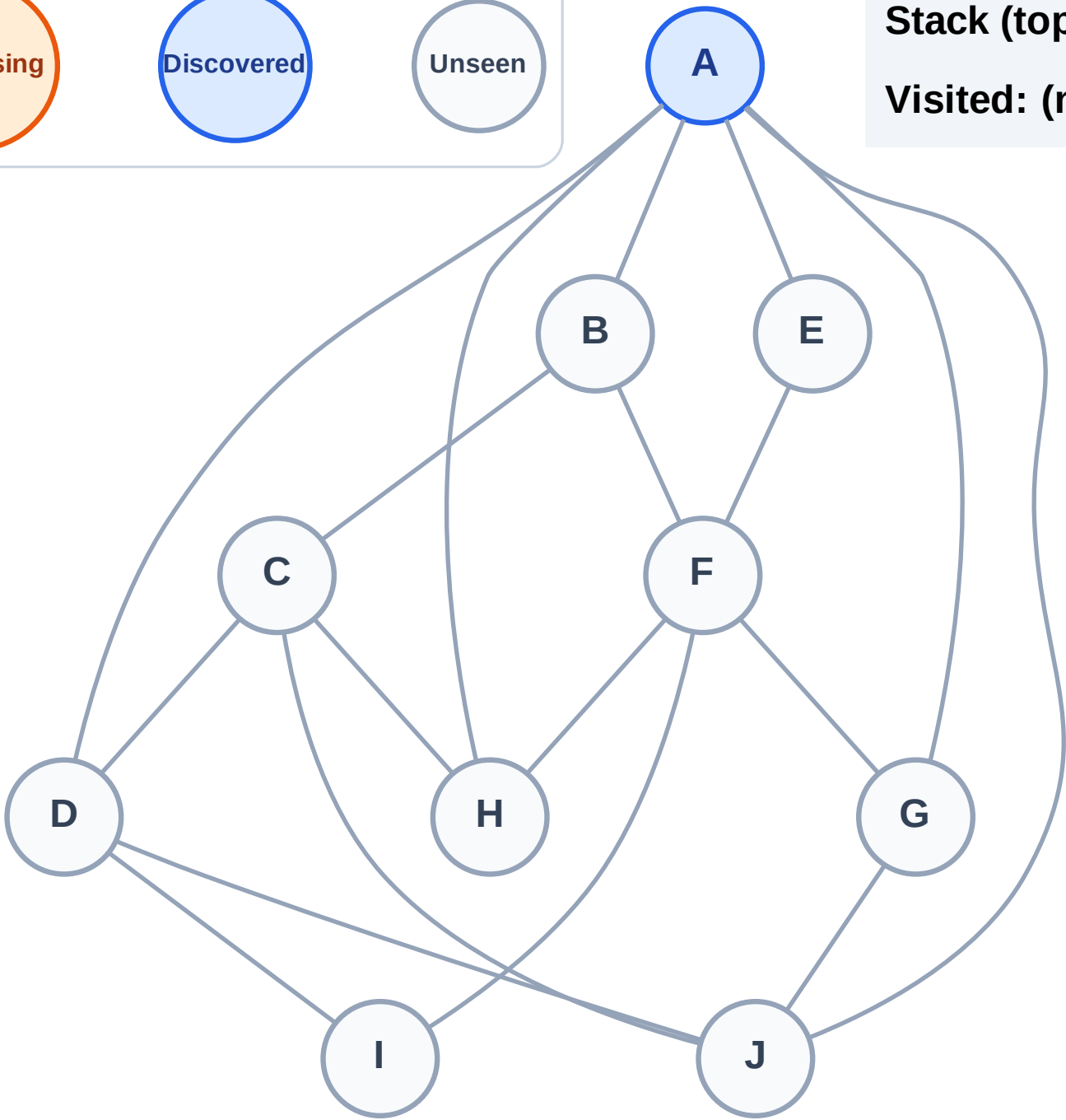
Step 1: Start at A

Legend



Stack (top at right): A

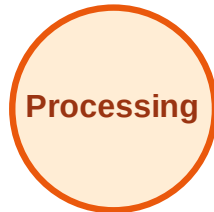
Visited: (none)



DFS Traversal

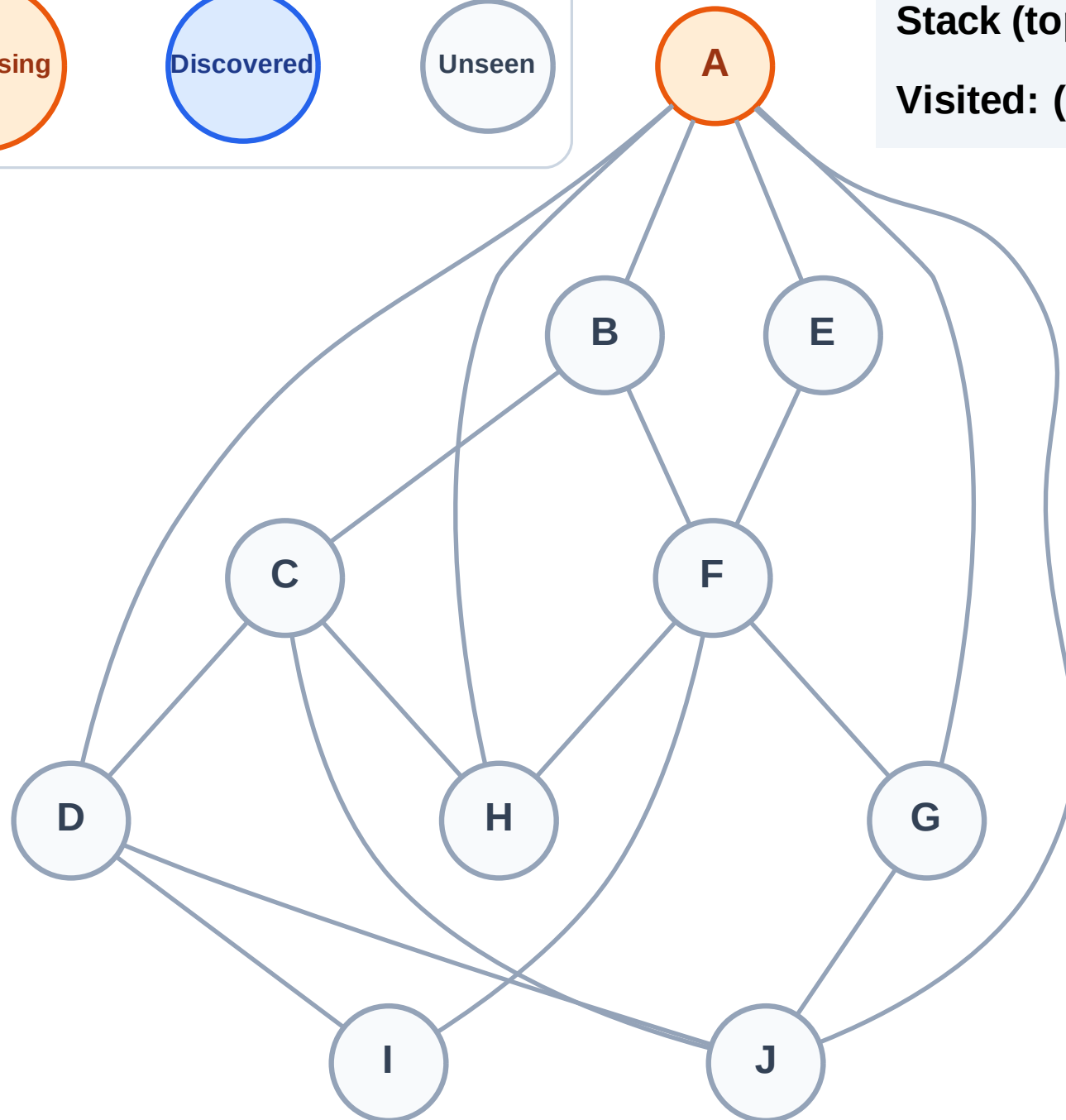
Step 2: Process node A

Legend



Stack (top at right): (empty)

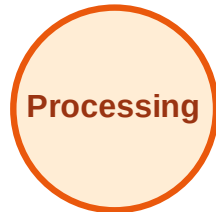
Visited: (none)



DFS Traversal

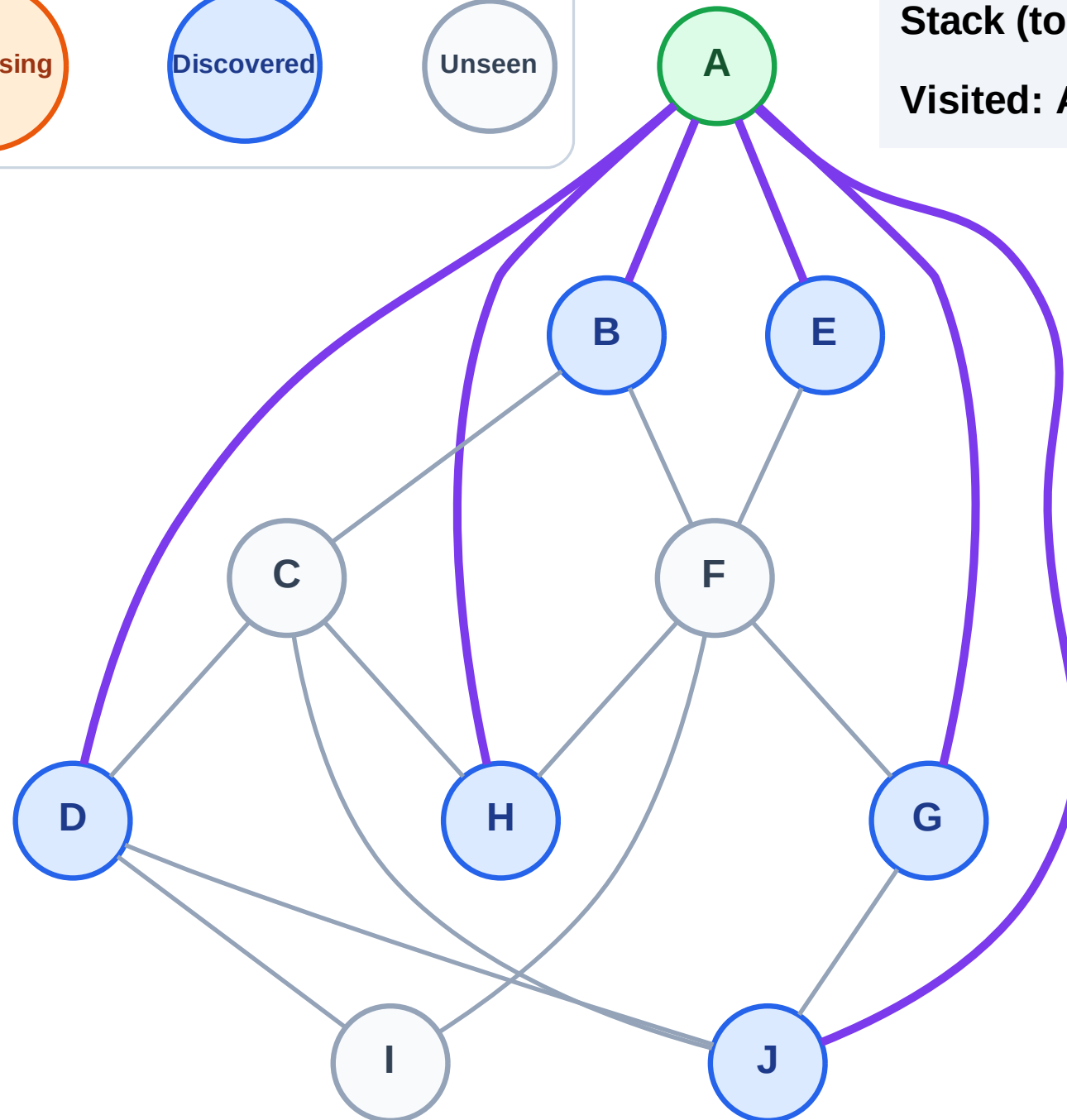
Step 3: Discover J, H, G, E, D, B from A

Legend



Stack (top at right): J | H | G | E | D | B

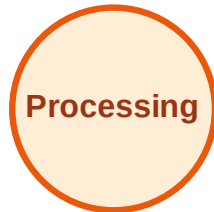
Visited: A



DFS Traversal

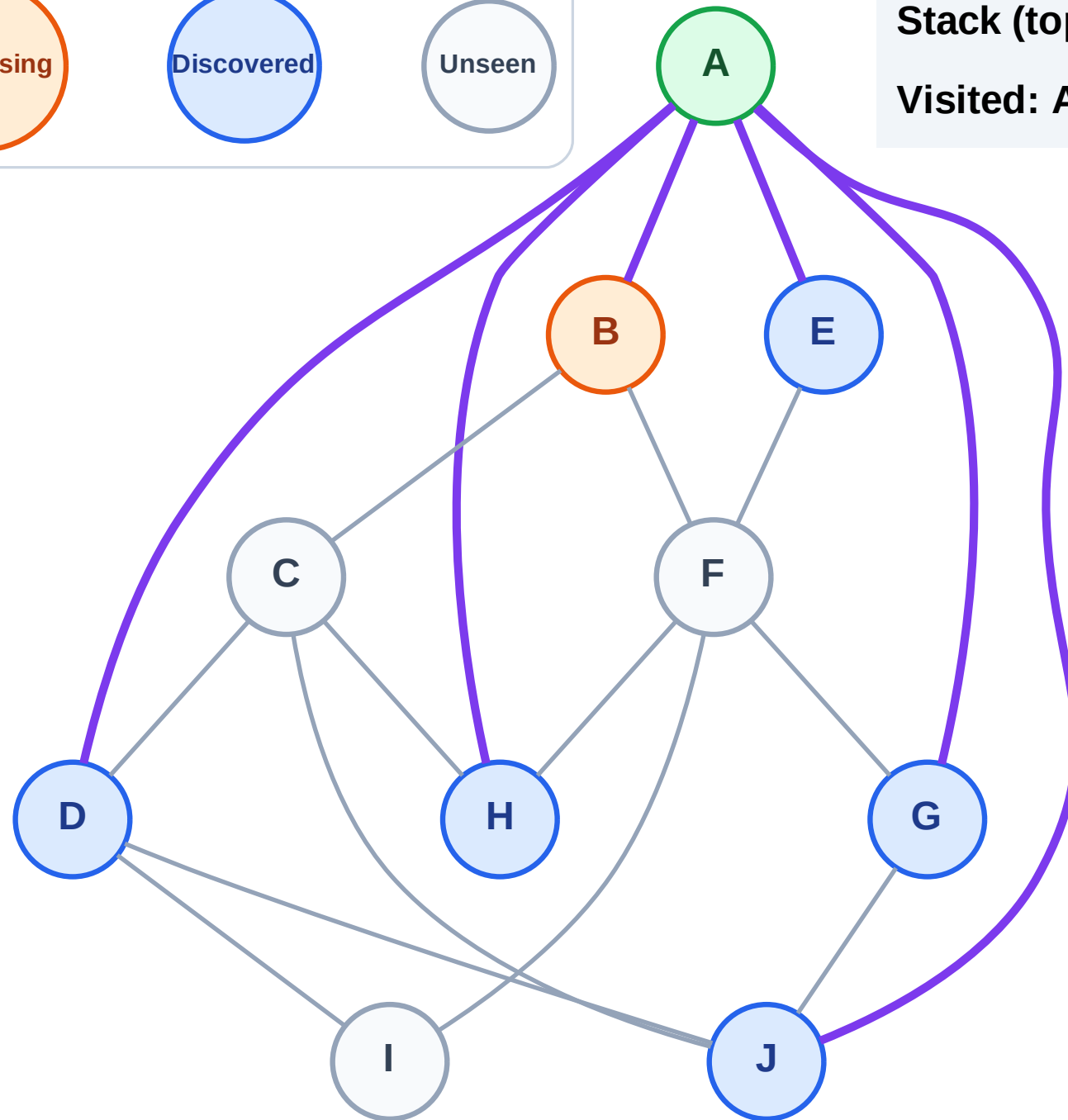
Step 4: Process node B

Legend



Stack (top at right): J | H | G | E | D

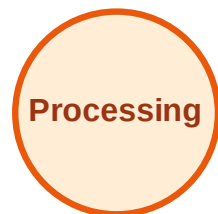
Visited: A



DFS Traversal

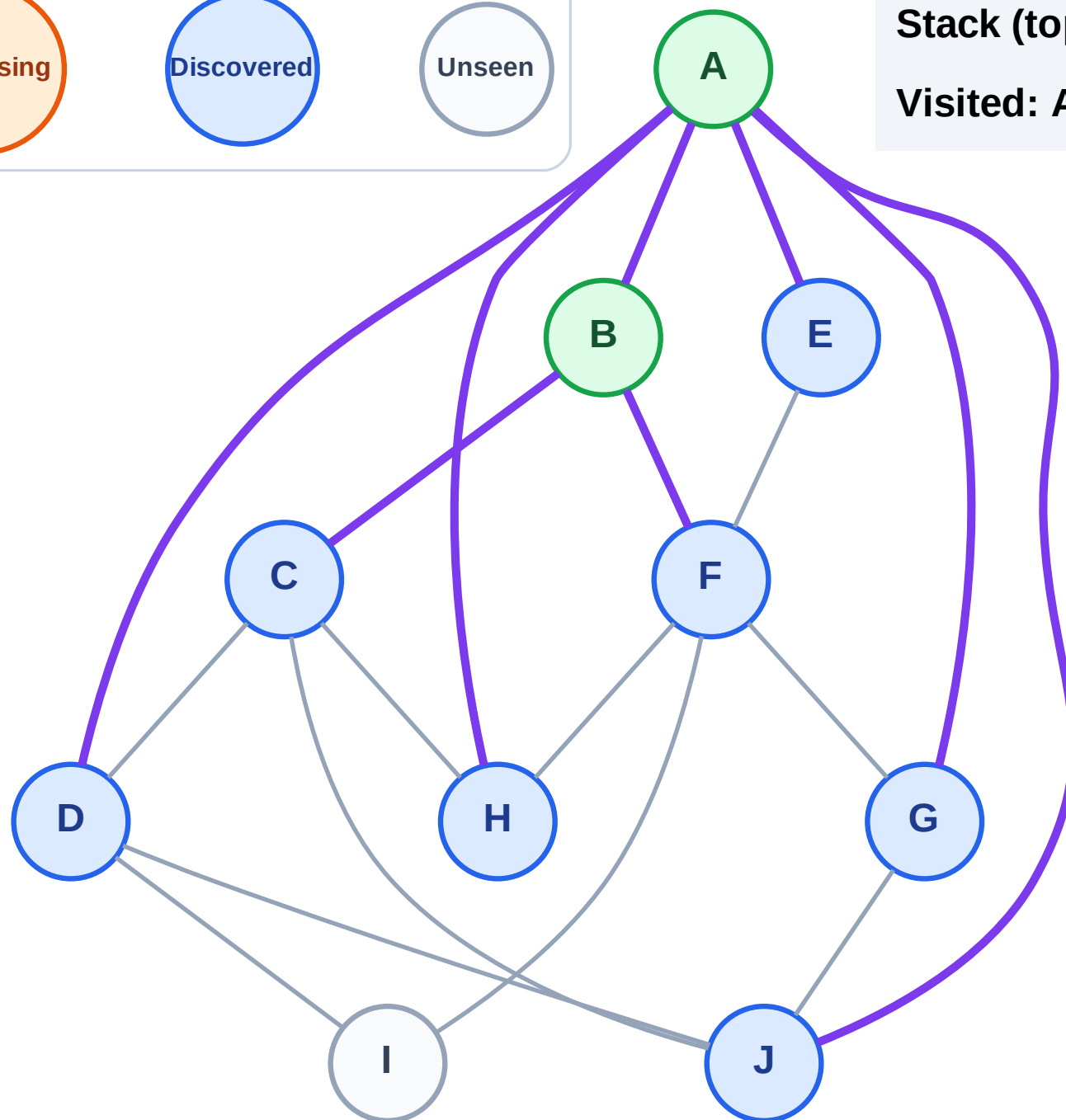
Step 5: Discover F, C from B

Legend



Stack (top at right): J | H | G | E | D | F | C

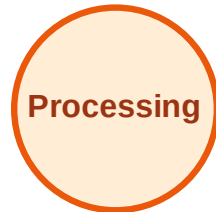
Visited: A, B



DFS Traversal

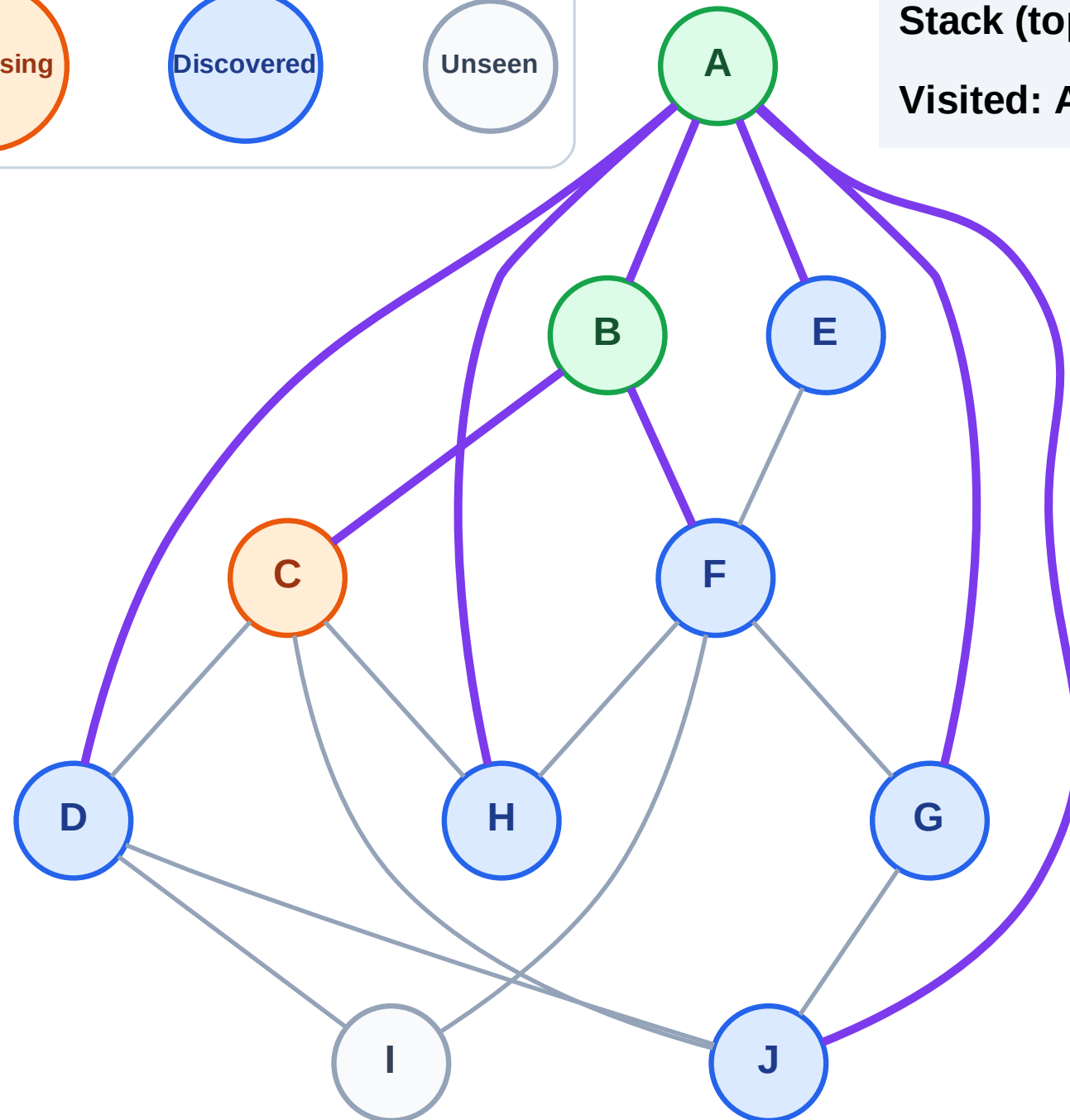
Step 6: Process node C

Legend



Stack (top at right): J | H | G | E | D | F

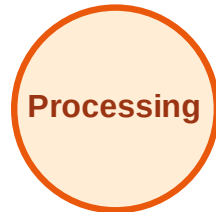
Visited: A, B



DFS Traversal

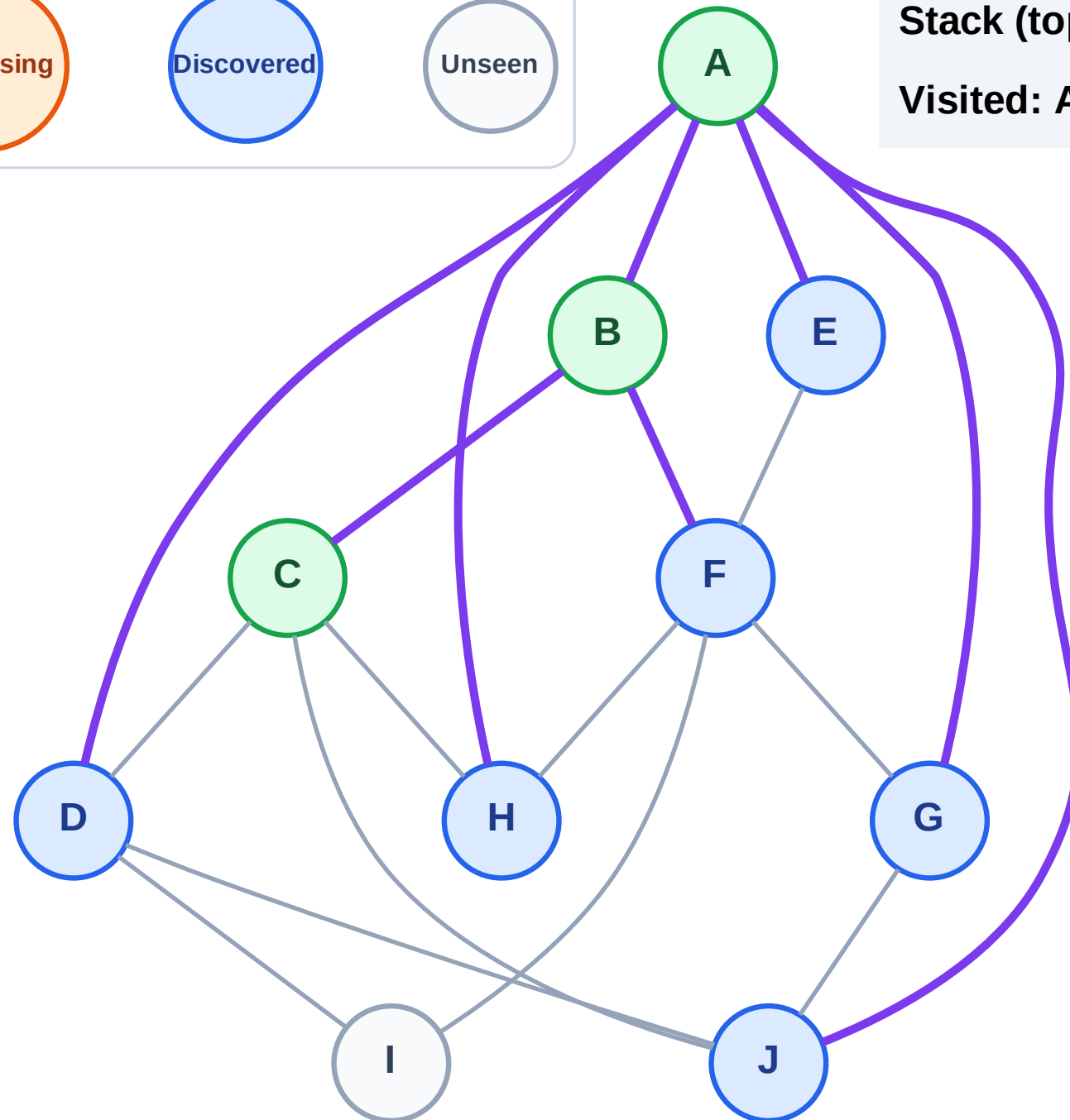
Step 7: No new neighbors from C

Legend



Stack (top at right): J | H | G | E | D | F

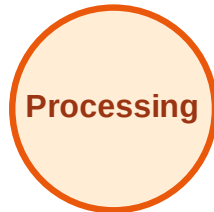
Visited: A, B, C



DFS Traversal

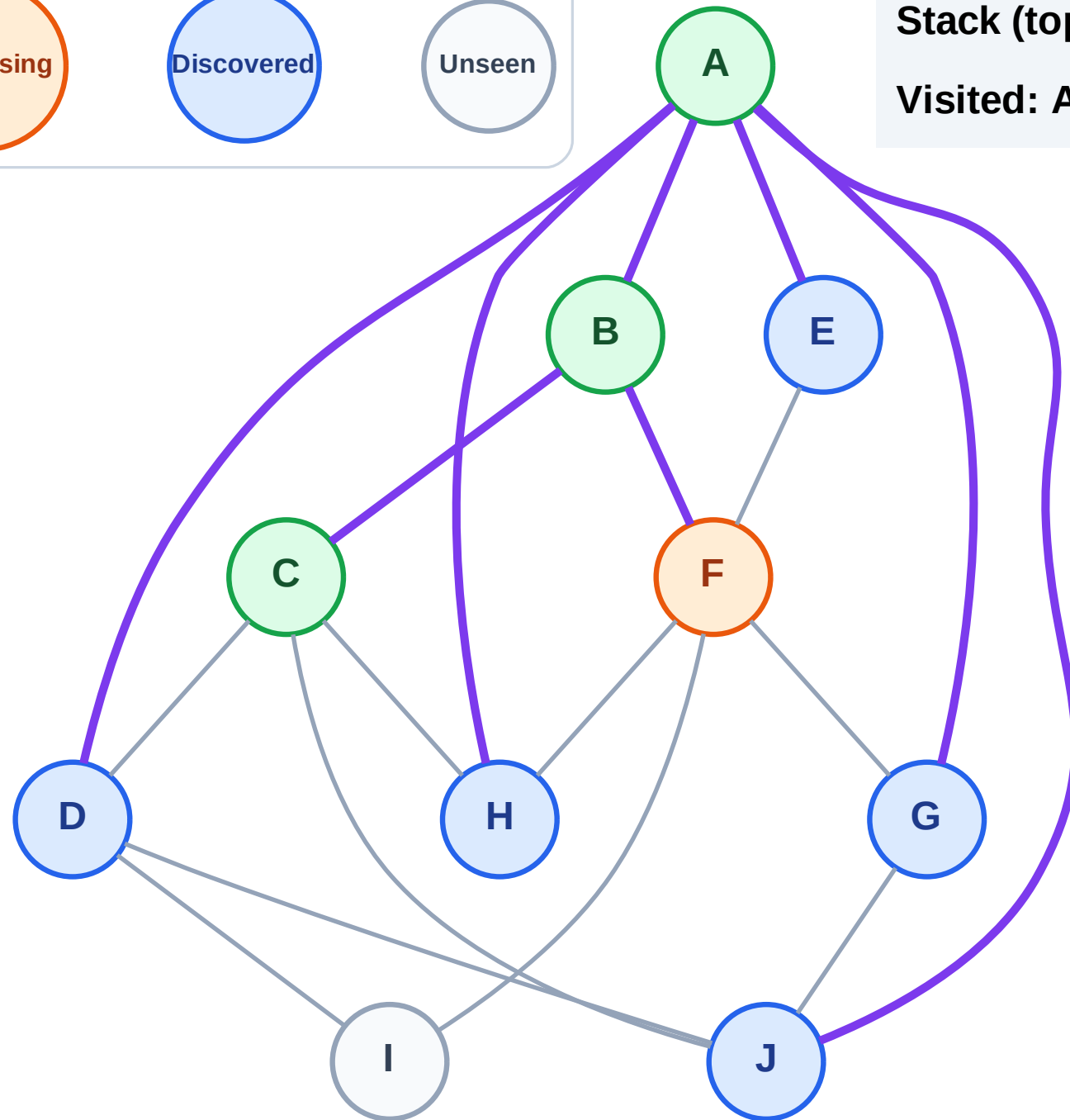
Step 8: Process node F

Legend



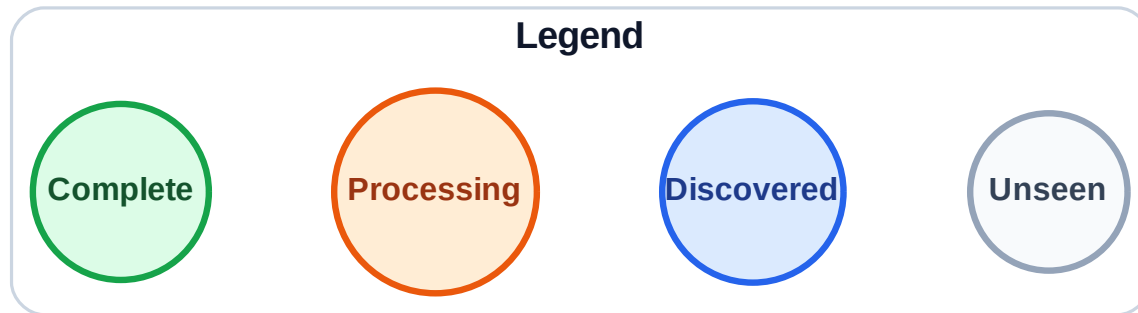
Stack (top at right): J | H | G | E | D

Visited: A, B, C



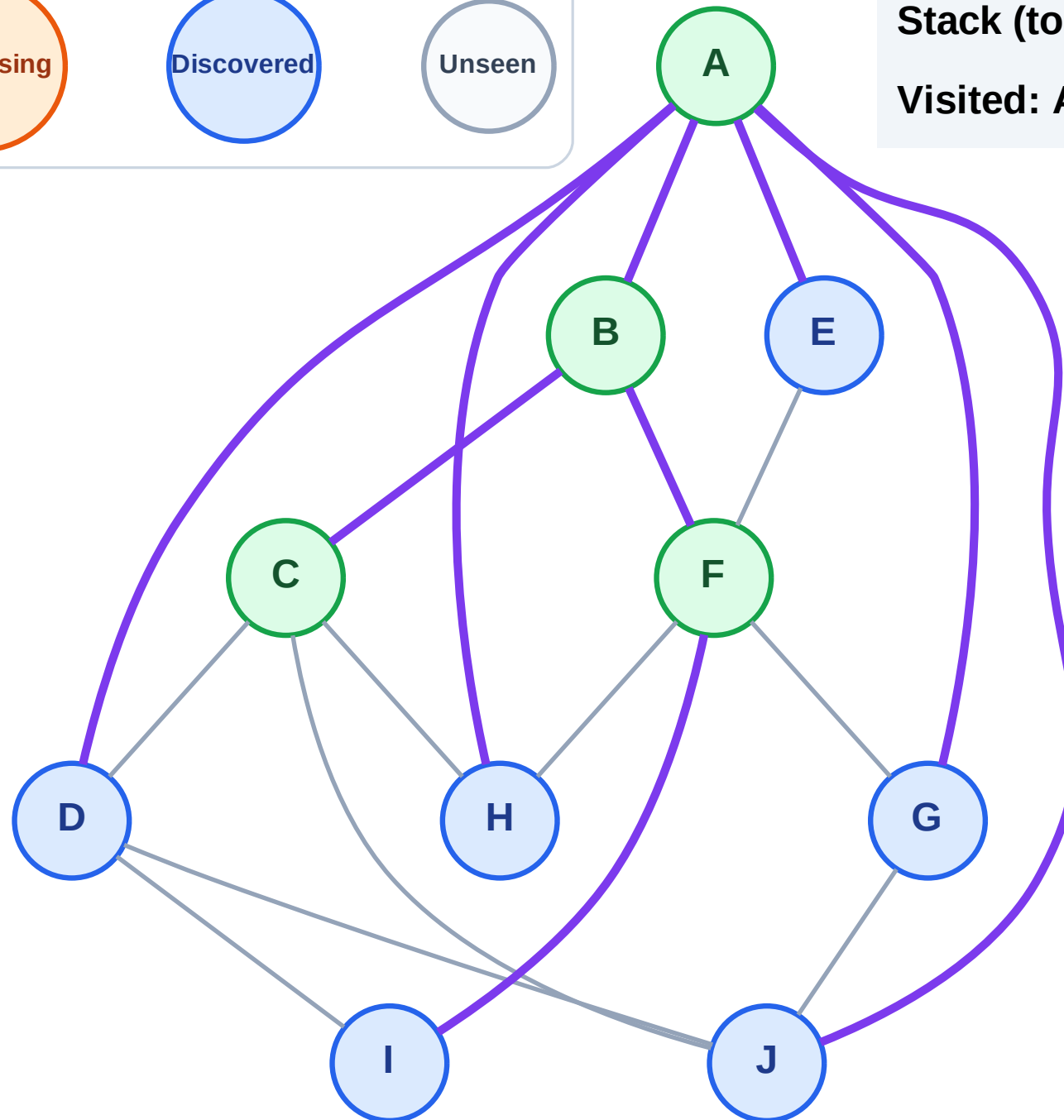
Step 9: Discover I from F

Step 9: Discover I from F



Stack (top at right): J | H | G | E | D | I

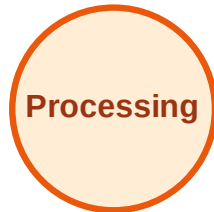
Visited: A, B, C, F



DFS Traversal

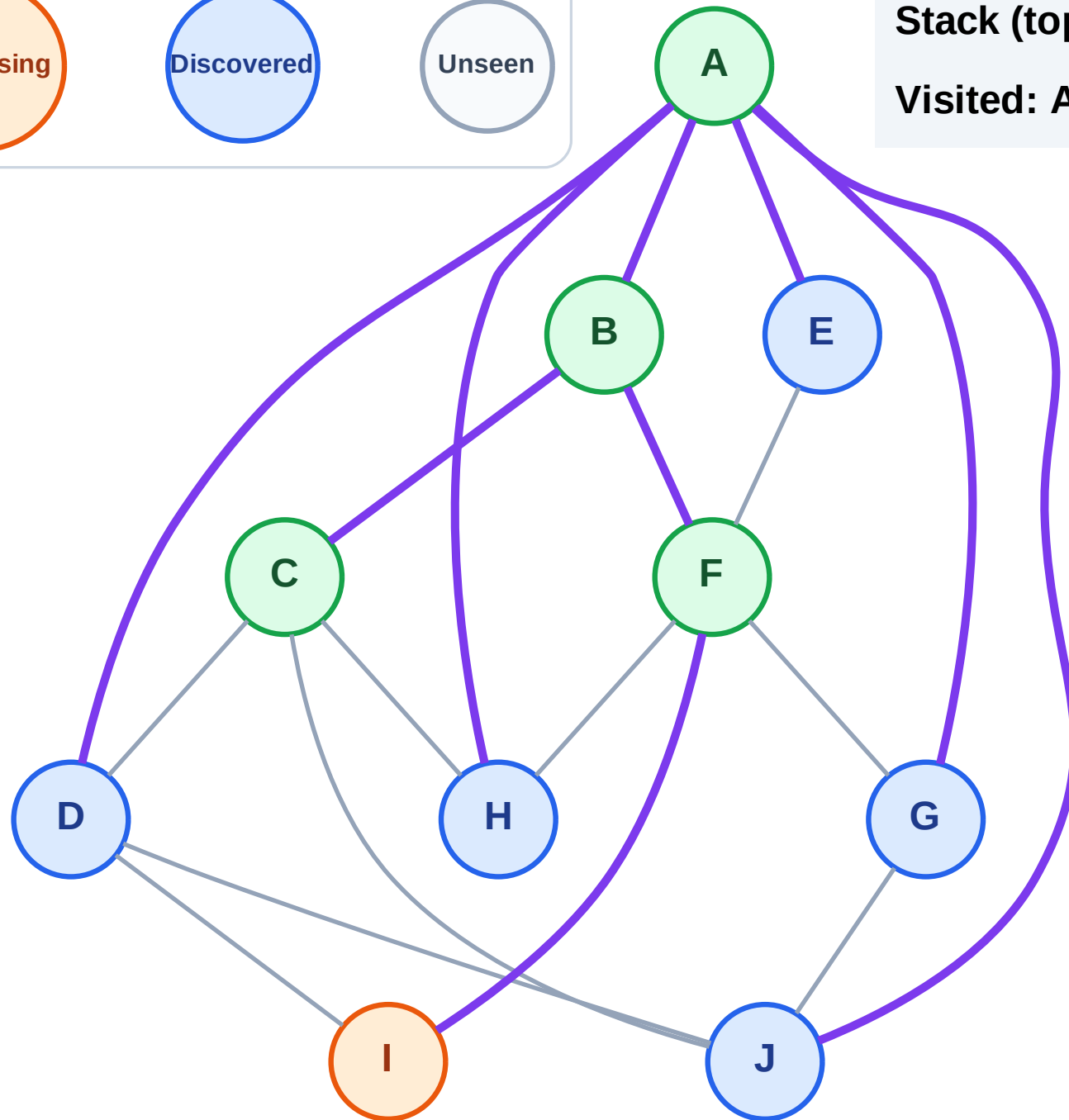
Step 10: Process node I

Legend



Stack (top at right): J | H | G | E | D

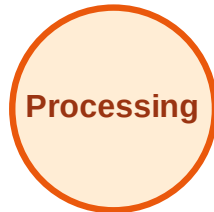
Visited: A, B, C, F



DFS Traversal

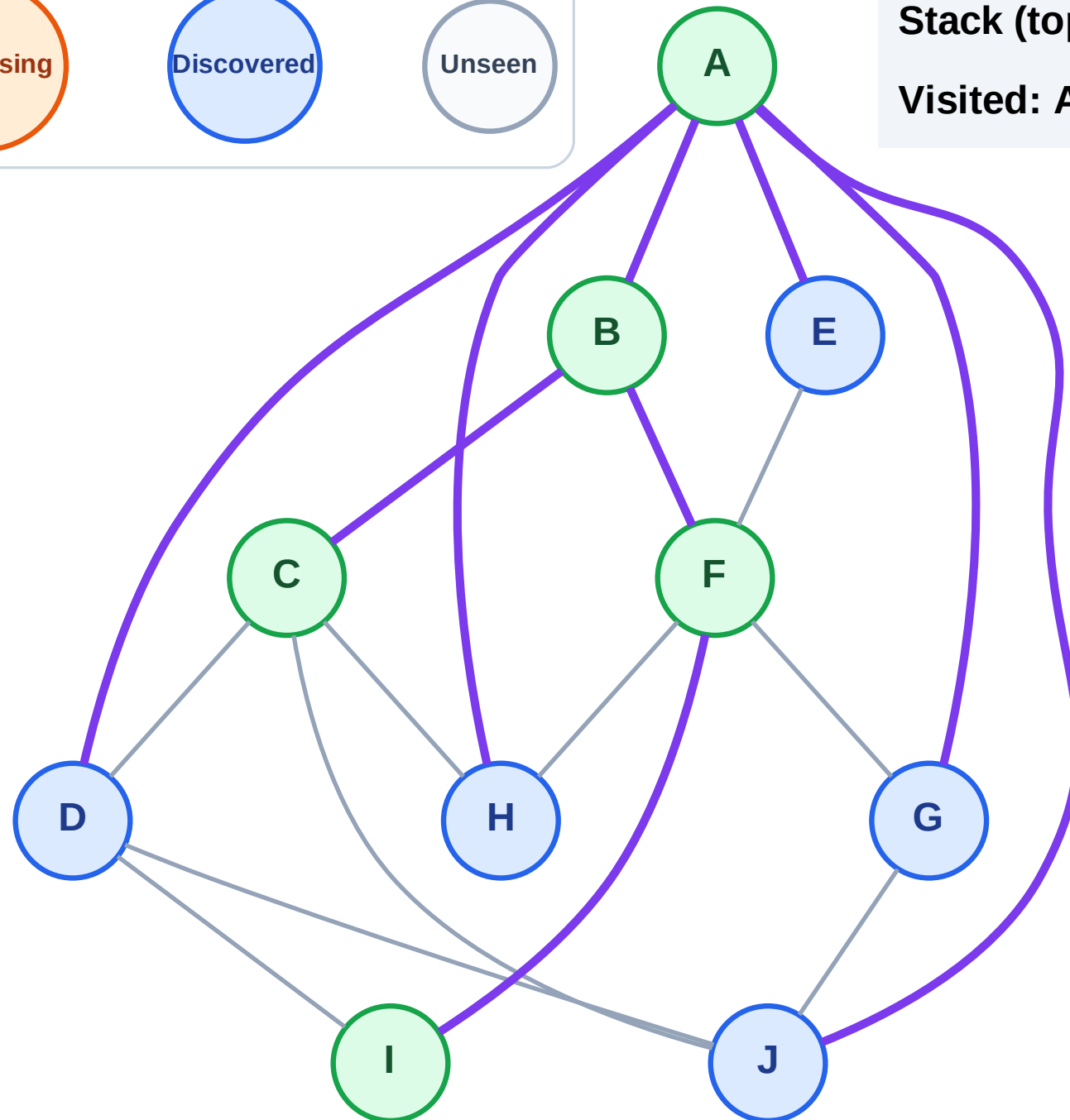
Step 11: No new neighbors from I

Legend



Stack (top at right): J | H | G | E | D

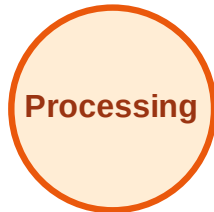
Visited: A, B, C, F, I



DFS Traversal

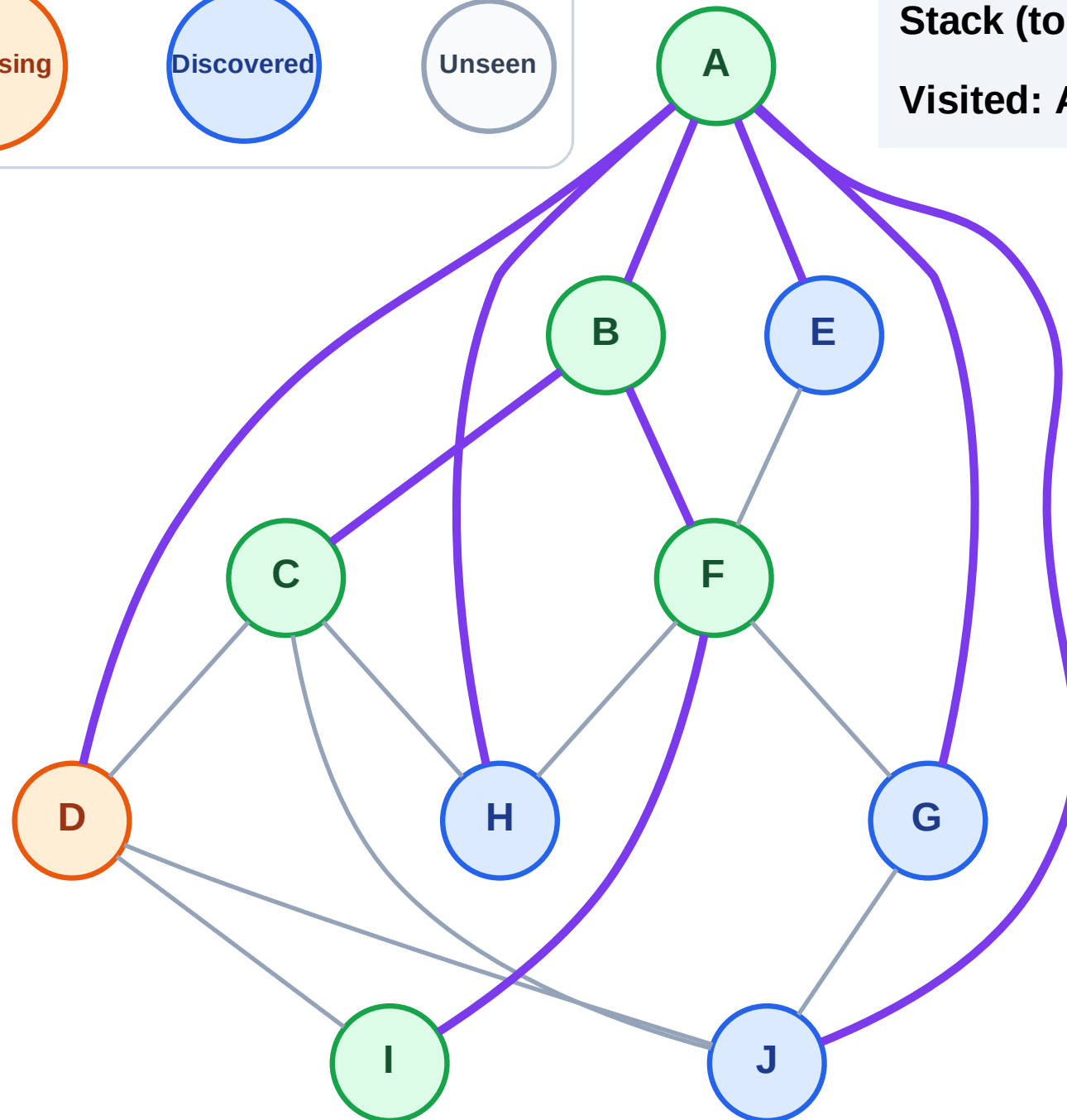
Step 12: Process node D

Legend



Stack (top at right): J | H | G | E

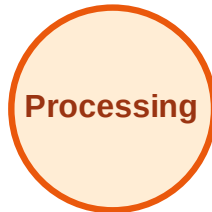
Visited: A, B, C, F, I



DFS Traversal

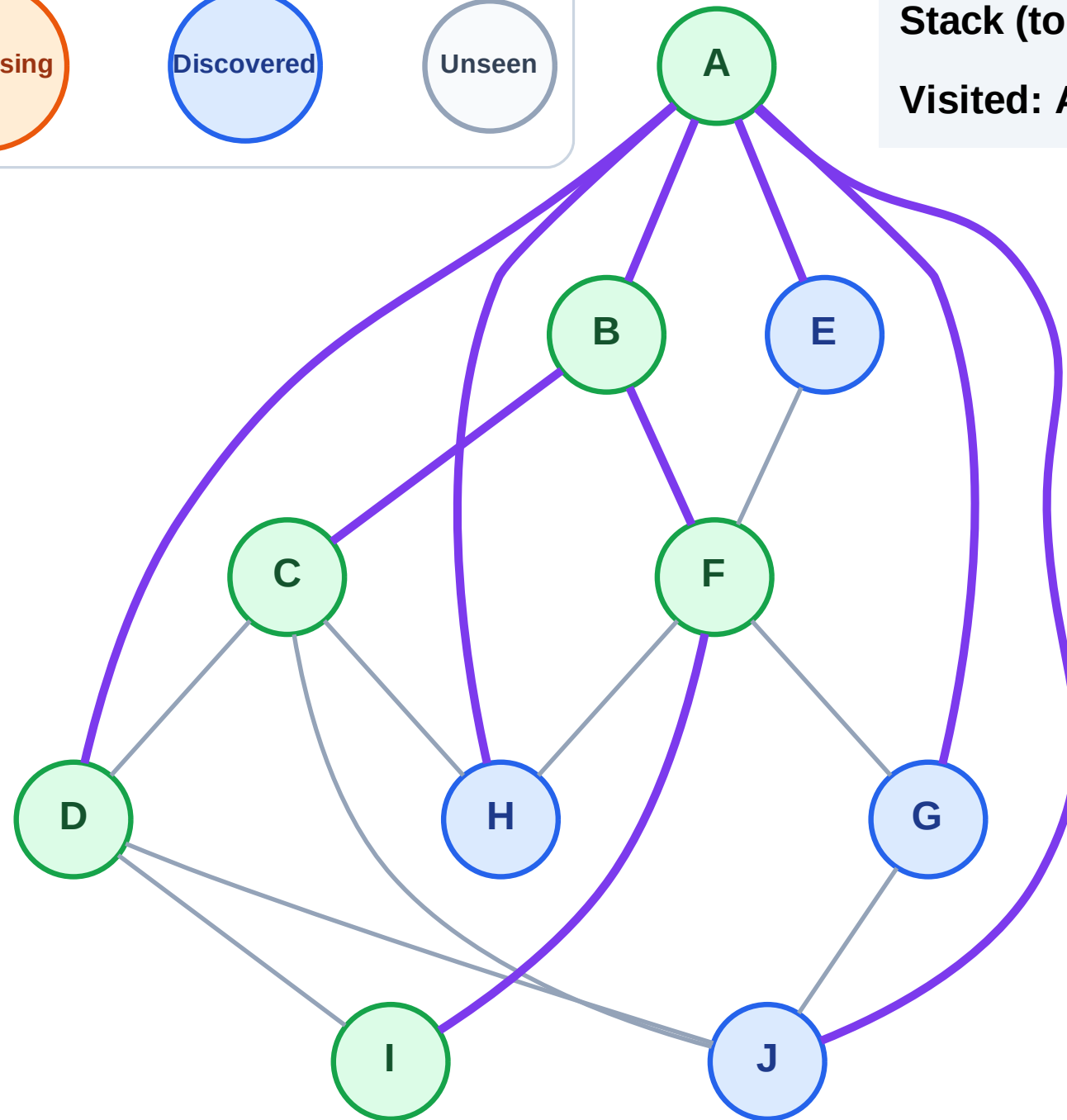
Step 13: No new neighbors from D

Legend



Stack (top at right): J | H | G | E

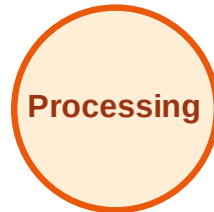
Visited: A, B, C, D, F, I



DFS Traversal

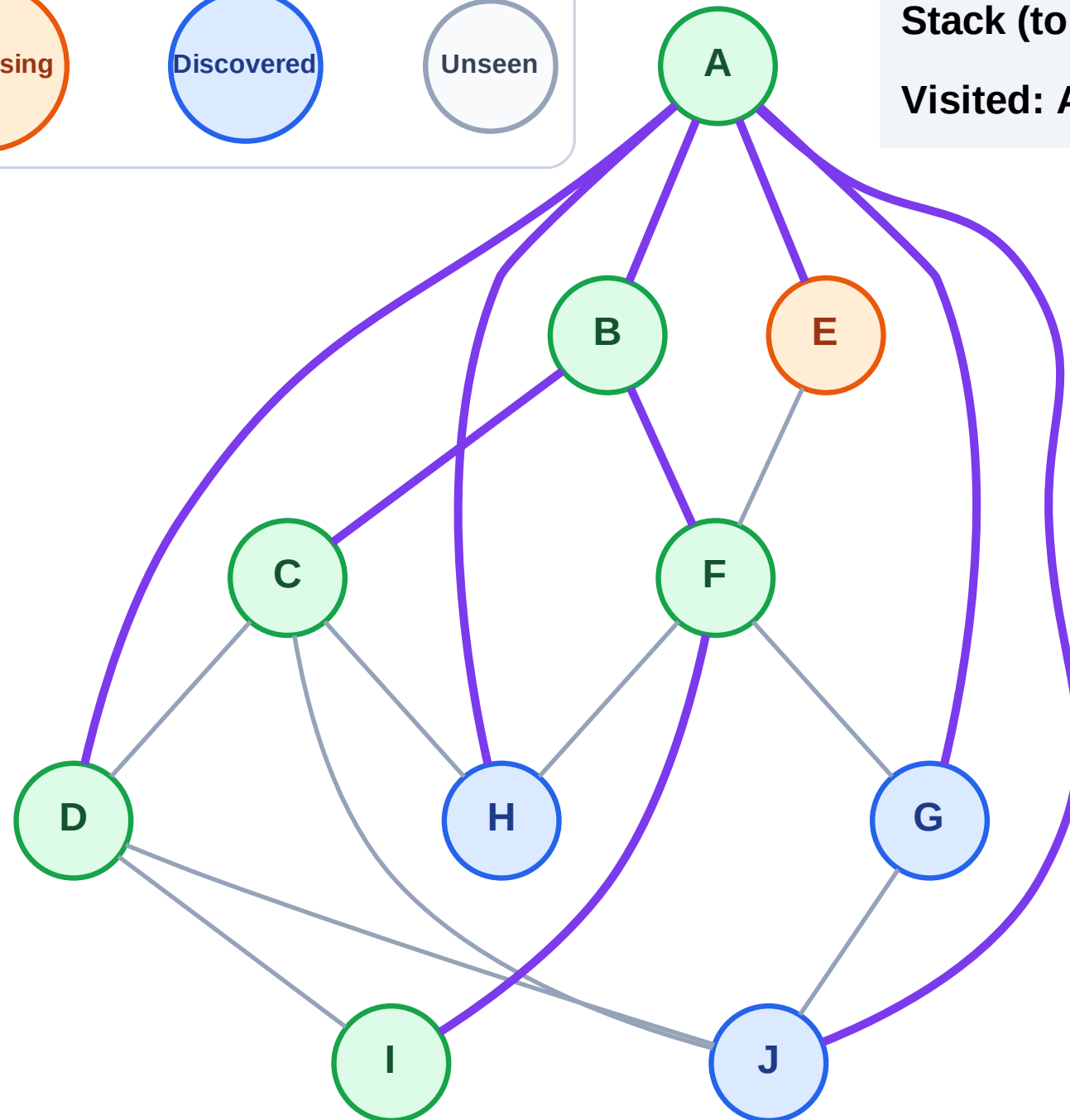
Step 14: Process node E

Legend



Stack (top at right): J | H | G

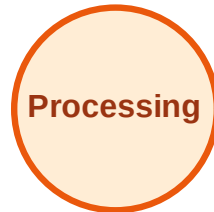
Visited: A, B, C, D, F, I



DFS Traversal

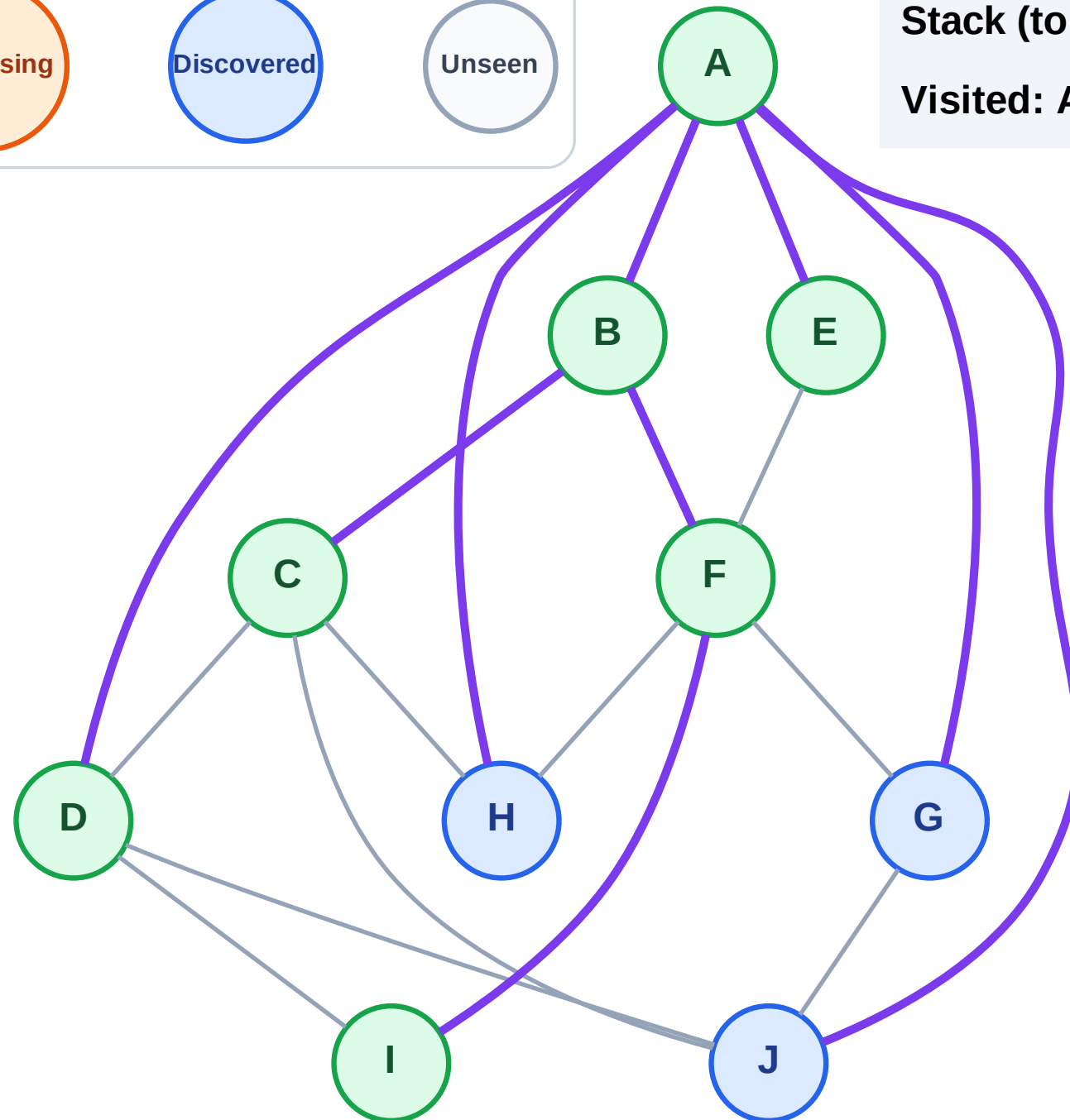
Step 15: No new neighbors from E

Legend



Stack (top at right): J | H | G

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 16: Process node G

Legend

Complete

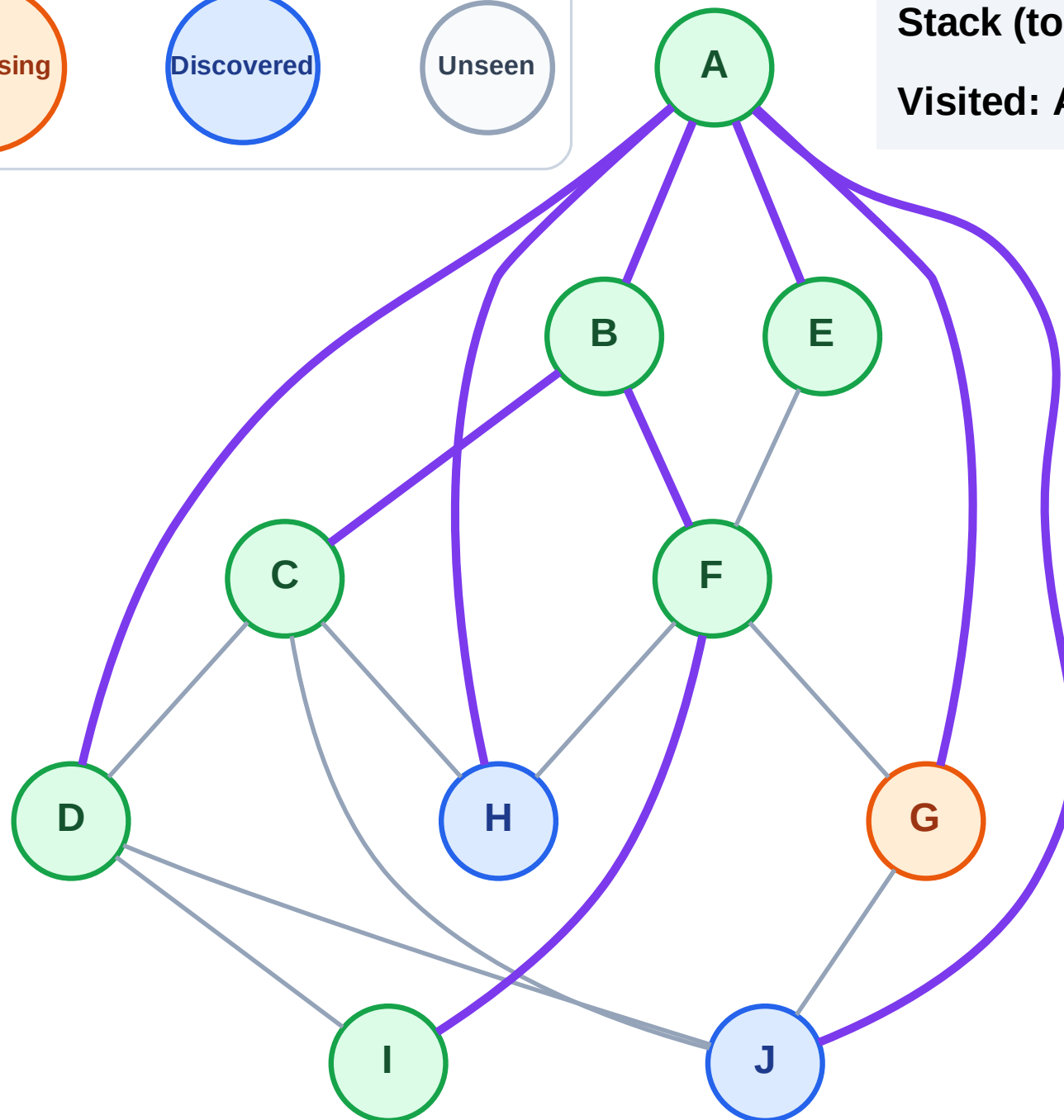
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 17: No new neighbors from G

Legend

Complete

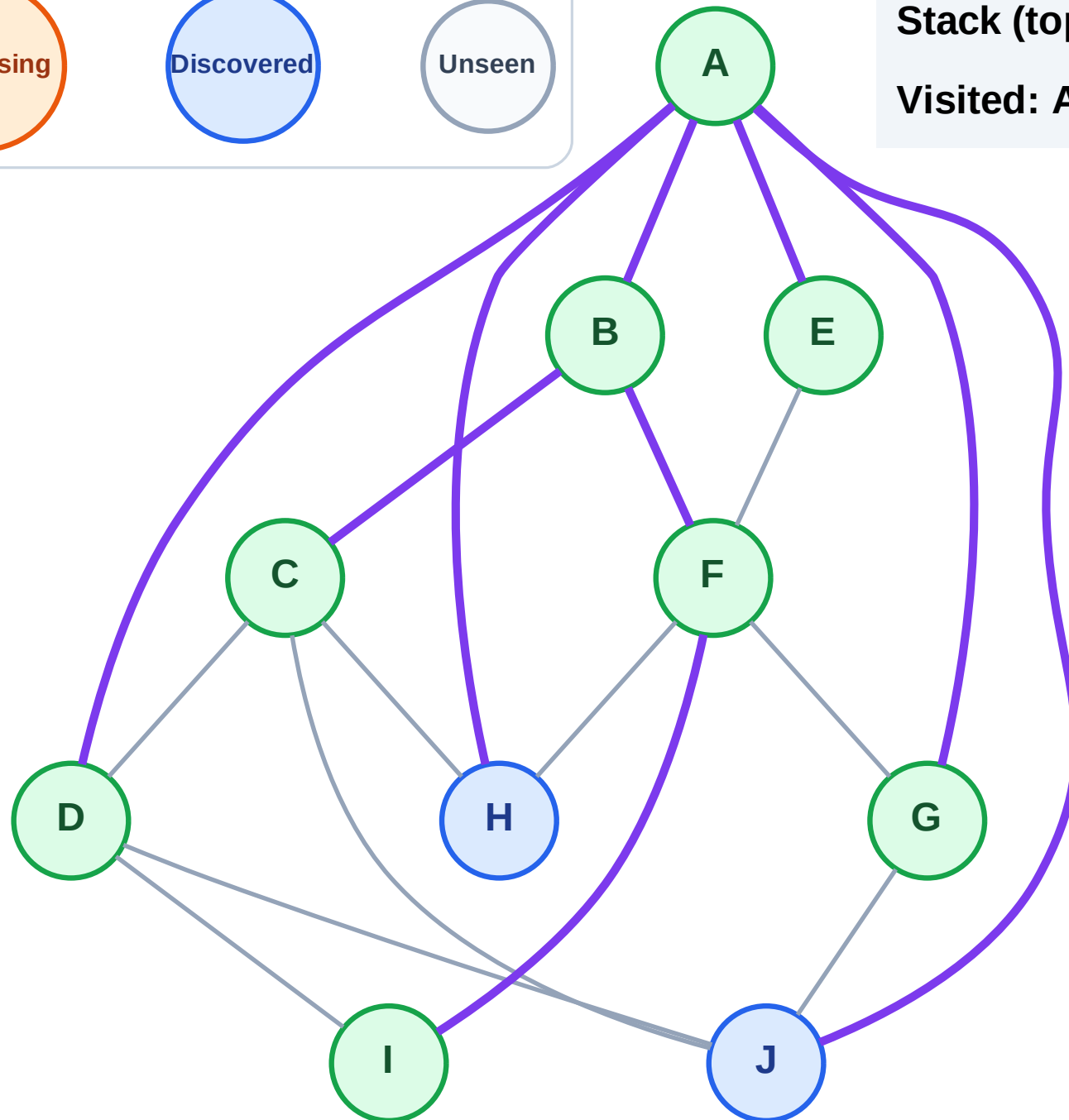
Processing

Discovered

Unseen

Stack (top at right): J | H

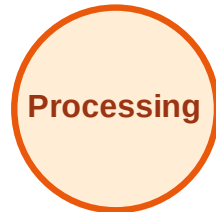
Visited: A, B, C, D, E, F, G, I



DFS Traversal

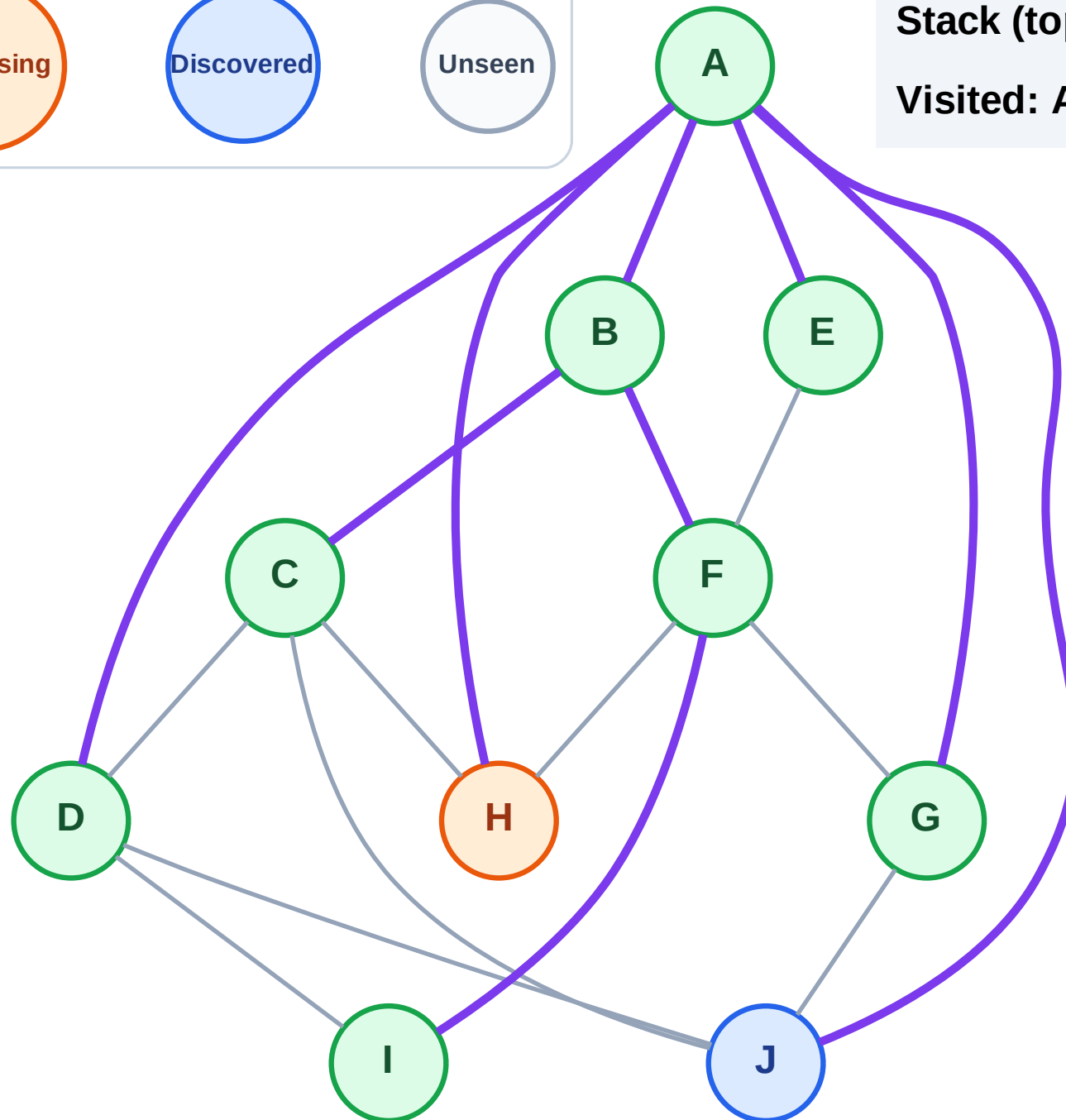
Step 18: Process node H

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, I

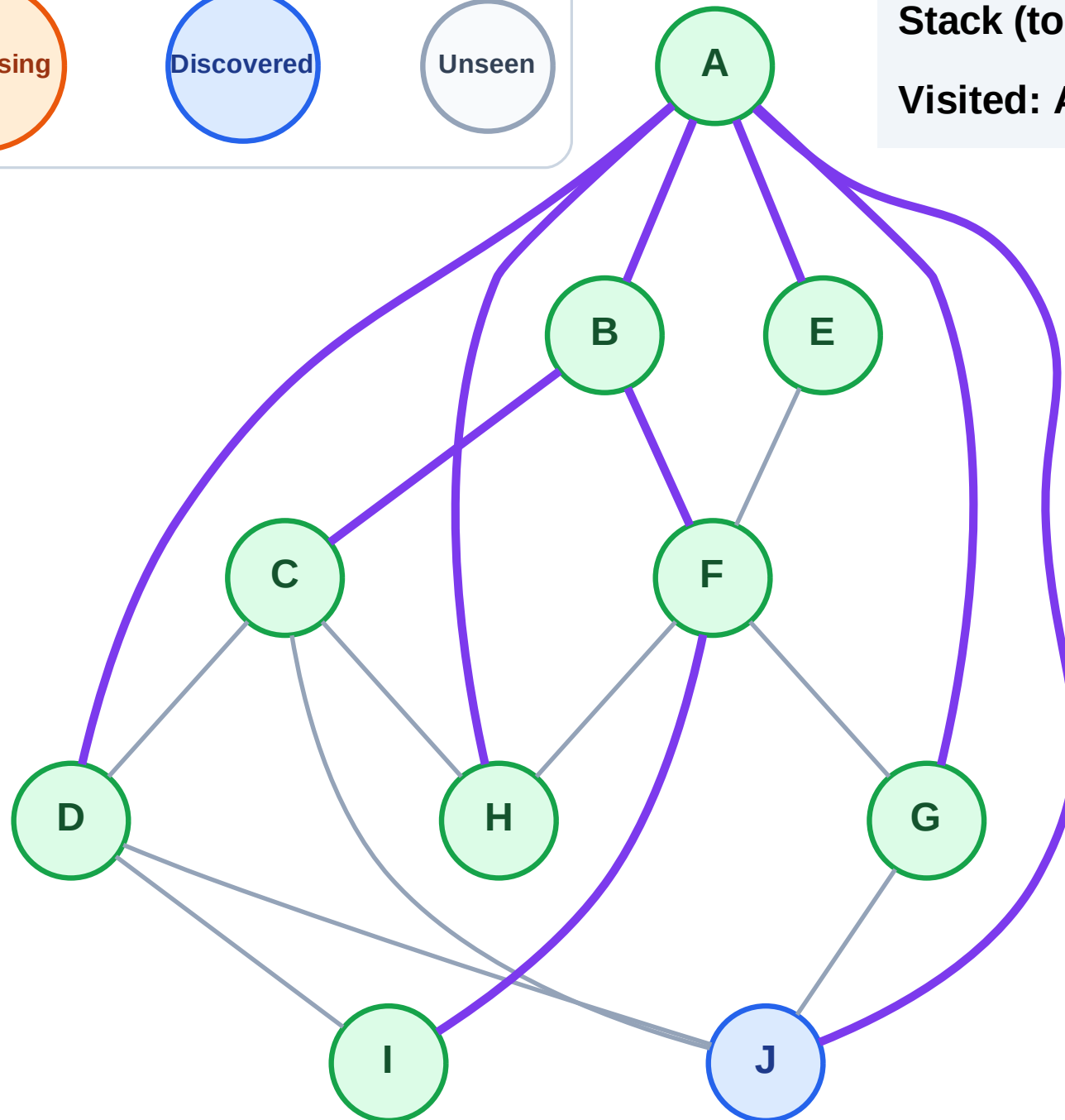


Step 19: No new neighbors from H

Legend



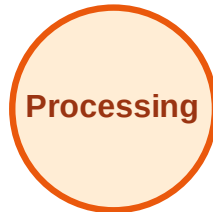
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

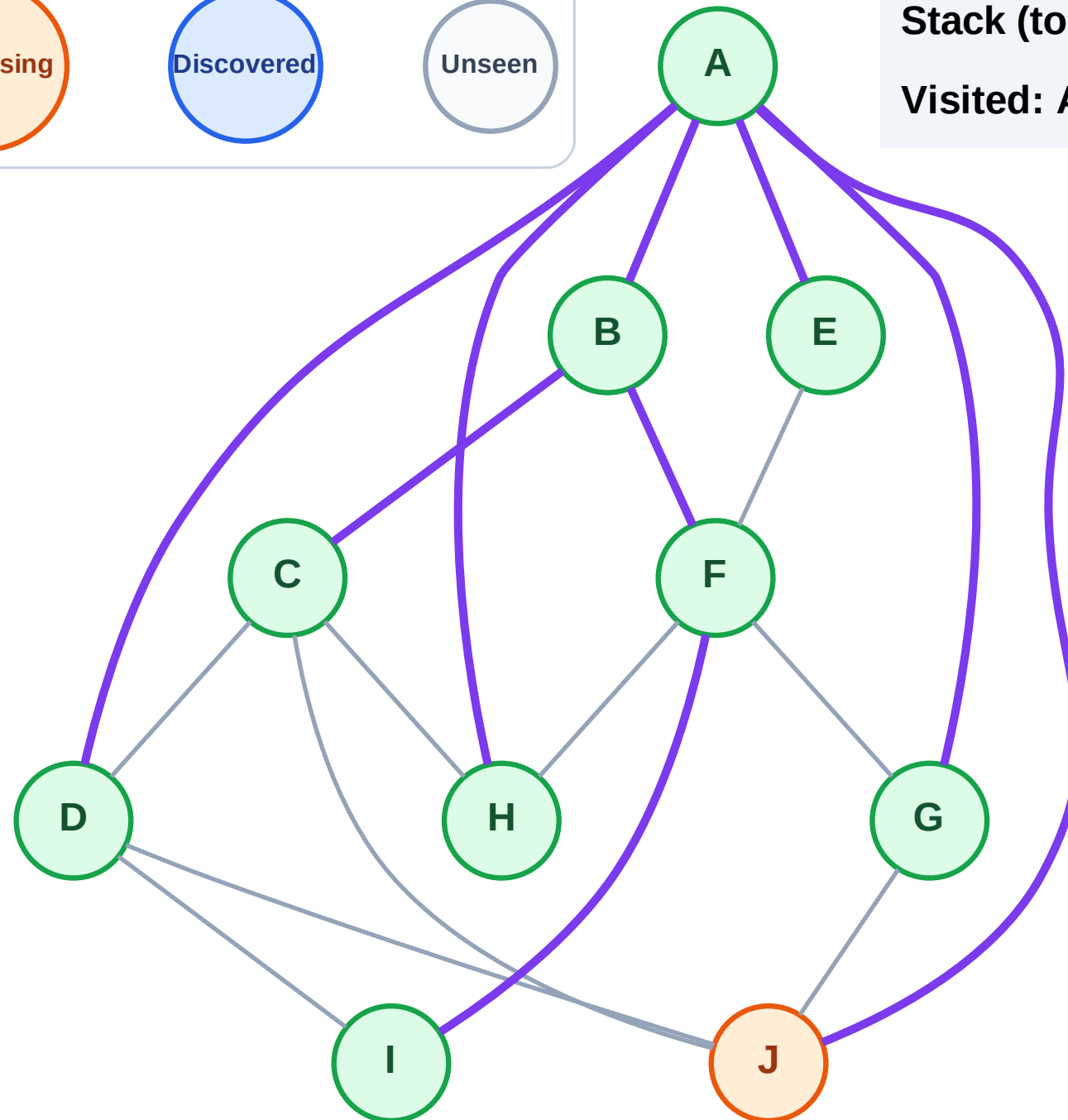
Step 20: Process node J

Legend



Stack (top at right): (empty)

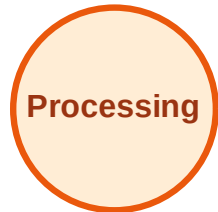
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

